Formalizing Microentrepreneurs through Targeted Tax Systems: Evidence from Brazil

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Abstract

We study the effects of reducing formalization costs by exploiting the introduction of a simplified tax regime for microentrepreneurs in Brazil. Using novel administrative data and industry-level eligibility criteria, we show that a large reduction in entry and compliance costs significantly increased formal firm creation. Despite the increase in formalization, newly formal firms exhibit no growth or earnings gains. We conclude by studying the program's fiscal consequences and show that, while it successfully expanded access to social security benefits among microentrepreneurs, the associated public expenditures were 2.56 times greater than the revenue it generated through tax collections.

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1 Introduction

Most businesses in developing countries are small and informal (Jayachandran, 2020). High levels of informality raise concerns among policymakers due to reduced tax revenues, limited access to social security, and the strong association between informality and lower productivity and earnings (Levy, 2008). As a result, there is significant interest in policies that encourage formalization. One widely discussed approach is to reduce the costs of formalization. While appealing in theory, the empirical evidence on the effectiveness of this strategy remains mixed (Ulyssea, 2020), and debates persist over whether the key barrier is the fixed cost of entry or the ongoing costs of remaining formal. Despite this attention, less is known about the fiscal consequences of such policies—especially the trade-off between lower tax revenues and increased public spending through expanded social security coverage of formalized individuals.

In this paper, we study the effects of one of the largest formalization programs in the world, leveraging novel administrative records and household survey data that capture both formal and informal labor markets. The program, called *Microempreendedor Individual* (MEI), was launched in Brazil in July 2009 and introduced a special tax regime that substantially reduced formalization costs for eligible microentrepreneurs. It simplified bookkeeping requirements, lowered tax rates, and eliminated registration fees. Prior to the program, the costs of registering a formal firm were estimated at 2,000 BRL—roughly four times the minimum wage—and required about 120 days. After full implementation, eligible individuals could register under MEI in a single day and at virtually no cost.

Our analysis focuses on three core dimensions. First, we examine how the reduction in formalization costs affected firm registration and the number of active formal firms among microentrepreneurs, and use the phased rollout of the program to shed light on the relative importance of different mechanisms—namely, entry costs versus tax rates. Second, we document a substantial increase in formalization among microentrepreneurs, but find no corresponding improvements in earnings or firm growth. Third, we evaluate the fiscal implications of the policy by quantifying both the increase in tax revenue and the additional costs associated with expanded social security coverage.

We begin our analysis by examining the effects of the MEI program on formal firm entry and activity. To identify causal effects, we exploit the program's industry-level eligibility criteria, defined at the 7-digit industry classification (CNAE). Our empirical strategy compares outcomes for eligible and ineligible industries before and after the program's implementation using a difference-in-differences design. We construct a monthly panel of industries using comprehensive firm-level administrative data from the national business registry (CNPJ), which allows us to track firm creation, and closure for all formal businesses over time.[1]

[1] Our data differs from previous work that relied on employer-employee matched data and only observed microentrepreneurs with formal employees. Rocha et al. (2018), uses RAIS to estimate the effects of the MEI program but severely underestimate its impact on formal firm registration and activity, as MEI firms without employees are not required to report to RAIS. We find that fewer than 5% of MEI firm owners hire a formal

We find that the MEI program led to a large increase in firm registration and the number of active formal firms. Our estimates indicate that the program caused the number of active formal firms to increase by 86% in eligible industries by December 2015. These effects are driven entirely by microentrepreneurs entering the MEI tax system, with no change in the number of formal firms outside the MEI regime. The timing and sharpness of these effects are consistent with a causal interpretation. Our granular administrative data allows us to document the parallel trends prior to the program and to precisely attribute the observed growth to firms entering under the MEI regime.

We use the phased structure of the MEI program to investigate the mechanisms driving these effects. In its initial phase, beginning in July 2009, the program simplified taxes and bookkeeping requirements but retained substantial registration costs. In February 2010, a second phase introduced a fully integrated online registration system that eliminated in-person bureaucracy and virtually all entry costs. Finally, in May 2011, the government implemented a significant tax cut, reducing the flat monthly payment from 11% to 5% of the minimum wage. We compare responses across these phases to distinguish the effects of lowering entry barriers from those of reducing ongoing tax liabilities. Our findings indicate that the sharpest increase in firm registration occurred after the drop in registration costs, while the subsequent tax cut had no meaningful additional impact. This suggests that the burden of entry—rather than the level of taxation—was the primary constraint to formalization for most microentrepreneurs.

We then move beyond firm registration to better understand the program's impacts on formalization and microentrepreneurs' earnings, using household survey data that captures both formal and informal labor market activity. We draw on the nationally representative PNAD survey, which provides detailed information on individuals' employment status, social security contributions, and earnings. To implement our empirical strategy, we map PNAD's industry classifications to the program's eligibility rules and construct an industry-by-year panel. This allows us to compare trends in formalization and income across eligible and ineligible sectors over time using a difference-in-differences design.

We show that the program substantially increased formalization among microentrepreneurs but had no discernible effects on their earnings. Our difference-in-differences estimates indicate that, by 2015, the number of formal microentrepreneurs in eligible industries had grown by nearly 80% relative to ineligible industries. In contrast, average labor income among microentrepreneurs remained flat, with no significant differences between eligible and ineligible groups over time. At the aggregate level, the share of formal microentrepreneurs rose by 15 percentage points over six years—an increase of 79% from the baseline—while the total number of microentrepreneurs remained stable, indicating that the program induced formalization rather than shifts in occupational choice.

We complement these results with descriptive evidence from our administrative data, which allows us to track the full universe of MEI firms over time, including detailed information

worker.

on ownership, employment, and transitions into and out of formal labor market ties. First, we document that newly formalized microentrepreneurs exhibit minimal firm growth. Four years after registration, 97% of MEI firms remain nonemployers, and even among those that hire at least one worker, the 90th percentile employs no more than five workers. Second, we find no evidence that the observed increase in formalization is driven by individuals gaming the system—such as formally employed workers reclassifying themselves as MEI to reduce tax liabilities. Using linked employer-employee records, we show that the vast majority of MEI registrants were not previously employed in the formal sector, and transitions out of formal jobs at the time of MEI registration are limited and gradual.

The final part of our analysis assesses the fiscal implications of the MEI program. While policies that encourage formalization are often motivated by the prospect of increased tax collection, they may also expand access to contributory benefits, generating additional fiscal costs. Using administrative records on MEI contributions and benefit receipts from the Federal Social Security Agency (INSS), we quantify both the revenue raised through the program and the costs associated with new social security claims. Although the number of contributors grew to over 3.3 million by 2015, total annual revenue remained under 800 million BRL. This modest result reflects not only the program's deliberately low tax rate, but also limited compliance, with only about 55% of expected monthly contributions actually paid.

Despite the modest revenue gains, the program significantly expanded access to social security benefits among microentrepreneurs. We estimate that the number of benefit recipients in this group increased by 56% relative to other occupations over the same period. Combining these revenue and expenditure estimates, we find that the program generated costs approximately 2.56 times greater than the revenue it brought in, resulting in net fiscal loss of over 2 billion BRL in 2015.

This paper relates directly to the large literature that studies the causes of informality(La Porta and Shleifer, 2014). A central discussion in this literature has been about how registration costs and taxes affect informality. In a recent literature review, Ulyssea (2020) argues that reducing the costs of entering the formal sector has little to no effect on formalization, while reducing the costs of remaining in the formal sector can be more effective.

Our findings provide a counterpoint to this view. We show that a large simplification and reduction in entry costs produced substantial increases in formal firm registration, while a subsequent and sizable tax cut had no additional impact. These results highlight the salience of entry costs—particularly bureaucratic and time-related burdens—as a primary barrier to formalization among microentrepreneurs. This is consistent with evidence from Peru and Benin (Mullainathan and Schnabl, 2010; Benhassine et al., 2018), and helps reconcile mixed findings in the literature by emphasizing the importance of the magnitude of the cost reduction. In our context, the MEI program reduced registration time from 120 days to 1 and eliminated monetary costs, whereas studies that find smaller effects—such as in Sri Lanka (De Mel et al., 2013) and

Mexico (Bruhn, 2011, 2013)—focused on more modest reductions in administrative burdens.[2]

We also contribute to a broader debate about the value of formalization as a development strategy. A long tradition views informality as a key barrier to growth and allocative efficiency (De Soto, 1989; Busso et al., 2012; Dix-Carneiro et al., 2021; Zárate, 2019). However, a growing empirical literature finds limited economic gains from formalization at the micro level (De Mel et al., 2013; Benhassine et al., 2018). Our results are consistent with this view: we find that formalized microentrepreneurs rarely expand, hire workers, or increase earnings. This aligns with theoretical models in which reducing formality costs leads to higher formalization rates but minimal productivity or growth effects (Ulyssea, 2010, 2018).

An understudied dimension of formalization policies is their effect on access to social security coverage. While much of the literature focuses on firm behavior and tax compliance, much less is known about how these policies expand entitlement to contributory benefits. This gap is particularly relevant in light of Levy (2008), who emphasizes how segmented welfare systems in Latin America reinforce labor market dualism. We provide direct evidence on this channel by showing that the MEI program substantially increased access to pensions and other benefits among microentrepreneurs. However, the associated fiscal cost—driven by low contribution rates and a subsidized tax structure—was more than twice the revenue collected. These findings underscore the need to evaluate formalization policies not only through their effects on firm behavior, but also through their fiscal and redistributive implications.

This paper is the first to provide a comprehensive evaluation of the MEI program using administrative data that fully captures its target population. Rocha et al. (2018) rely on employer-employee matched data, which exclude the vast majority of MEI firms that operate without employees, and are therefore limited in their ability to shed light on the actual mechanisms driving formalization. By using complete administrative records of firm registration and social security contributions, we are able to capture the full extent of the program's effects on formalization, earnings, and benefit coverage. Two subsequent papers also study the MEI program. Pereda et al. (2022) focuses on gender differentials using household survey data and finds women respond more than men to incentives, a pattern we also find in our data. Carvalho (2024) examines the welfare implications of the MEI program through a structural lens, adopting the same administrative data infrastructure and identification strategy introduced in this paper. However, their analysis does not examine underlying mechanisms, omits social security benefits from the fiscal evaluation, and does not use administrative data to document firm growth or transitions between employment and MEI status—resulting in a more limited treatment of informality and fiscal consequences of the program.

[2] The SIMPLES tax system in Brazil has also promoted a debate on the effects of taxes on informality and development (Monteiro and Assunção, 2012; Fajnzylber et al., 2011), but recent work shows that conclusions are often taken from insufficient data (Piza, 2018).

2 Institutional Setting and Data

As in many developing countries, microentrepreneurs constitute a large share of the Brazilian workforce. Defined as self-employed individuals or employers with at most one employee, they represented 22% of all workers aged 18 to 60 in 2008, according to the nationally representative household survey.[3] Yet only 17% of them contributed to the tax authority, compared to a 65% formalization rate among employees. Moreover, during the first five years of the *Partido dos Trabalhadores* (Workers' Party) federal administration, there was no decline in microentrepreneur informality. In contrast, employee formalization rose by seven percentage points over the same period.[4]

In December 2008, the Brazilian federal government launched the *Microempreendedor Individual* (MEI) program, which began operating in July 2009. The program aimed to increase the formalization rate of microentrepreneurs and, more broadly, to promote economic gains among newly formal firms. It introduced a new legal category within the tax system, allowing eligible microentrepreneurs to register as MEI firms. Registered firms gained access to the same formal benefits as other legal entities, including the ability to issue invoices and receipts and obtain formal credit. MEI registration also brought firms into compliance with government regulations. In addition, firm owners became eligible for contributory benefits such as maternity leave, disability insurance, and survivor pensions.

Eligibility Criteria: To register as a MEI firm, microentrepreneurs must meet four criteria: (i) The owner cannot be a partner or shareholder in any other firm, and each MEI firm must have only one owner; (ii) Annual revenue must be below a threshold, originally set at 36,000 BRL; (iii) The firm may employ at most one worker; (iv) The firm must operate in one of the industries deemed eligible by regulation.

Industries are defined according to the *Classificação Nacional de Atividades Econômicas* (CNAE), a classification system maintained by the Brazilian Institute of Geography and Statistics (IBGE) and also used for tax records. Each CNAE code is a seven-digit number: the first two digits represent a broad sector, the next three provide a narrower classification, and the final two define the most specific activity.[5] This structure enables us to control for aggregate sectoral trends using interactions between leading CNAE digits and period fixed effects.

The list of MEI-eligible industries was derived from the SIMPLES tax system, Brazil's first major effort to reduce the tax burden on small firms. Eligible activities are concentrated in lower value-added sectors such as small-scale retail, construction, and low-skilled services.[6] While

[3] *Pesquisa Nacional por Amostra Domiciliar* (PNAD), conducted annually by the Brazilian Institute of Geography and Statistics. See below for details.

[4] The Workers' Party frequently advocated for labor formalization. Reducing informality was an explicit government objective in official platforms during the 2002 and 2006 presidential campaigns.

[5] For example, 4721-1/02 is the CNAE code for bakeries. 47 refers to all retail activities, 472 to retail of perishable goods, 4721-1 to retail of bread, milk, and similar products, and 4721-1/02 specifically to bakeries.

[6] Eligibility is defined by Resolution No. 58 of the *Conselho de Gestão do Simples Nacional*.

eligible and ineligible industries differ, we are not aware of any contemporaneous, sector-specific shocks that would confound our identification strategy. We describe our identifying assumptions in detail in Section 3 and show that there are no differential pre-trends in formalization between eligible and ineligible sectors.

Changes in Formalization Costs: While MEI firms were entitled to the same legal and institutional benefits as other formal firms, the costs of formality were substantially lower for those registered under the MEI tax regime. These cost reductions occurred along three key dimensions: entry costs, taxes, and bookkeeping requirements.

Before the MEI program, the lowest-cost option available to a microentrepreneur seeking formalization was the SIMPLES tax system.[7] SIMPLES features tax brackets that vary by industry and firm revenue. In what follows, we compare the costs faced by MEI firms to those of SIMPLES firms operating in the services sector within the same revenue range. These differences—across taxes, entry procedures, and compliance requirements—are discussed below and summarized in Table 1.

Taxes: MEI firms pay a flat tax calculated as a percentage of the minimum wage. From the program's launch until April 2011, the tax rate was 11% of the minimum wage; beginning in May 2011, it was reduced to 5%. In contrast, comparable firms in the SIMPLES tax system paid a rate of 4% on their gross revenue.

Because the two systems are based on different tax bases—minimum wage versus revenue—it is not straightforward to compare their effective tax burdens.[8] Nonetheless, we can approximate the relative burden of MEI taxation using household survey data. In 2009, the minimum wage was 465 BRL, implying a monthly MEI tax of 51.15 BRL. The median earnings of informal microentrepreneurs at the time were 600 BRL,[9] suggesting an implicit tax rate of 8.5% on profits. By 2011, when the tax rate was reduced, the minimum wage had risen to 545 BRL, lowering the flat monthly tax to 27.25 BRL. Median earnings of informal microentrepreneurs in 2011 were 760 BRL, implying an effective tax rate of approximately 3.5%. These figures indicate that the initial MEI tax rate was relatively high for a large share of microentrepreneurs, while the subsequent cut made MEI substantially more favorable—often lower than the equivalent SIMPLES rate.

Bookkeeping Costs: The flat tax structure of the MEI system substantially reduces bookkeeping requirements. MEI firms are not required to report the sources of their revenue or input costs and are exempt from submitting records reviewed by a certified accountant. In contrast, firms in the regular tax system must provide monthly accounting statements to the tax authority, a process that is often time-consuming and administratively burdensome. Regulations

[7] See Piza (2018) for a detailed overview of the SIMPLES regime.

[8] This challenge is compounded by data limitations. In surveys that include both formal and informal microentrepreneurs, we typically observe earnings (likely equivalent to profits), but not revenue. This prevents direct estimation of the elasticity of tax rates on outcomes like formal firm registration or informality.

[9] Author's calculations using PNAD 2009.

further require that these records be signed by an internal or external registered accountant. For microentrepreneurs, such services can be costly—sometimes exceeding the value of the taxes owed.

Registration Costs: Prior to the MEI program, registering a firm under the SIMPLES tax system required navigating a complex bureaucratic process involving federal, state, and municipal authorities. At the federal level, entrepreneurs had to register their firm with the national tax authority. At the state level, registration with the *Junta Comercial* (Trade Board) and the state tax authority was required. Municipalities added further requirements, including business licenses and, depending on the activity, approvals from the fire department, health agency, and environmental regulators. On average, this process cost around 2,000 BRL and took approximately 120 days to complete.[10]

When the MEI program began in July 2009, the registration process initially resembled that of other systems.[11] Individuals had to complete approximately 40 online forms, print and sign documents, and submit them in person to the Trade Board. This would initiate the state-level process, which proceeded until the firm was formally approved. Licenses and permits were only waived if local governments passed enabling legislation.

In February 2010, the federal government dramatically simplified the process. A new online platform allowed individuals to register by providing only their name, social security number, date of birth, and intended industry of operation.[12] Registration fees were eliminated, in-person appointments were no longer required, and the need for licenses and permits was removed in most cases. Firms could now complete the process in a single day and become immediately compliant with tax and regulatory obligations.

Program Timeline Summary: Although registration for MEI firms began in some states as early as July 2009, the program's implementation proceeded gradually. During the first phase (July 2009 to February 2010), states were progressively integrated into the system, but the unified online registration platform was not yet available. We define the second phase as the period from February 2010 to April 2011, when the fully integrated website was launched and made available nationwide, effectively eliminating registration costs. Throughout both phases, the MEI tax rate remained fixed at 11% of the minimum wage. In April 2011, the government reduced the tax rate to 5%, marking the beginning of the third phase, which extends through the end of our study period in December 2015.[13]

[10] See this report for a detailed breakdown of registration costs for non-MEI firms in 2009. Estimates are consistent with the World Bank's Doing Business survey (World Bank, 2010). Figure A1 summarizes the steps involved.

[11] Requirements varied by firm type. This presentation outlines the documents required to register a MEI firm in 2009.

[12] See this announcement from the federal government detailing the launch of the new system.

[13] Figure A2 summarizes the timeline of the MEI program.

Table 1: Comparing Cost for Firms in MEI and SIMPLES Systems

	MEI	**SIMPLES**
Tax Rate	Flat Tax Rate Jul/09 - Apr/11 - 11% of M.W. May/11 - onwards - 5% of M.W.	4% of Revenue
Entry Costs	Starting Feb/10 - Virtually Costless Integrated system in one Website Registration complete in 1 day	Avg. 2000 R$ process Different steps with at least 4 Government Agencies Average 120 days
Bookkeeping	No necessary Bookkeeping	Necessary Bookkeeping Accountant is mandatory by Law

This table plots the differences in cost of entry and compliance to firms in the formal sector under the MEI system and the previous best alternative to microentrepreneurs (SIMPLES). The SIMPLES tax system has different tax schemes according to industry and revenue progressivity. In this table compare the costs of a MEI firm relative to SIMPLES firm in the services sector in the same revenue range of MEI firms.

2.1 Data

We draw on two primary data sources for our analysis, supplemented by additional datasets described in Appendix A.

Cadastro Nacional de Pessoa Jurídica (CNPJ): The CNPJ is the national registry of all formal firms in Brazil, including those registered under the MEI program. It serves as the main building block of our dataset on formal microentrepreneurs. For each firm, we observe a unique firm identifier, official name, date of registration, date of closure (if applicable), industry classification (at the seven-digit CNAE level), legal nature of the entity,[14] address, and an indicator for whether the firm is registered under the MEI regime.

Importantly, for 93% of MEI firms, we can recover the owner's name and social security number (*Cadastro de Pessoa Física*, or CPF), which enables us to track unique microentrepreneurs.[15] Appendix A contains a detailed analysis of the structure and accuracy of the CNPJ data.

Pesquisa Nacional por Amostra Domiciliar (PNAD): Because the CNPJ data only captures formal firms, we rely on PNAD—a nationally representative household survey conducted

[14]In Brazil, all firms must declare their *Natureza Jurídica*, indicating whether they operate for profit or under another legal form.

[15]This is due to a design choice introduced in February 2010, when MEI firms' registered names (*Razão Social*) were required to include the owner's full name and CPF. Table A1 shows the share of firms for which we recover this information.

annually by IBGE—as our primary source for informal microentrepreneurs. PNAD contains individual-level information on demographics, employment, income, and social security contributions. We use it as a repeated cross-section, as individuals are not followed over time.

We prefer PNAD over the *Pesquisa Mensal de Emprego* (PME), which has been used in Rocha et al. (2018); Pereda et al. (2022); Carvalho (2024), for several reasons. First, PNAD provides a more detailed industry classification, with roughly 210 five-digit categories, and IBGE offers a direct crosswalk to the fiscal CNAE codes used to define MEI eligibility. In contrast, PME includes only 90 industry categories and lacks an official crosswalk. Second, PNAD is nationally representative, while PME covers only metropolitan areas. PNAD does have two limitations: it was not conducted in 2010 (a census year), and its annual frequency limits our ability to examine short-term variation. Nonetheless, in Section B, we replicate our key informality measures using PME data and find highly consistent results.

3 Effects on Formal Firm Entry and Activity

In this section, we examine the effects of the MEI program on formal firm registration and the number of active firms. We begin by outlining our empirical strategy, which exploits variation in industry-level eligibility. We then present the main results, explore potential mechanisms driving the observed effects, and address alternative explanations.

3.1 Empirical Strategy

To estimate the causal effect of the MEI program on firm entry and the number of active firms, we use a difference-in-differences approach that compares eligible and ineligible industries before and after the program's implementation. Industries are defined at the 7-digit CNAE level.

We aggregate firm-level data to this industry level and construct a monthly panel tracking firm entries, exits, and the stock of active firms. Treated industries are those eligible to register under the MEI regime. For our baseline estimates, we use all remaining industries as controls, and we conduct robustness checks using alternative control groups matched on pre-program characteristics. Our primary sample includes 1,323 unique CNAE industries, of which 371 are treated and 952 serve as controls.[16] The panel covers the period from January 2005 to December 2015.

In practice, we estimate the following difference-in-differences specification using this industry-month panel:

$$Y_{ct} = \theta_c + \gamma_{s(c)t} + \sum_{\substack{t=Jan2005 \\ t\neq June2009}}^{Dec2015} \beta_t \cdot D_{ct} + \epsilon_{ct} \tag{1}$$

[16]We exclude CNAE 9492800, which corresponds to political organizations, as electoral committees are legally required to register in this category during election periods.

where D_{ct} is a dummy that takes value one in month t if the 7-digit industry is eligible. We normalize the coefficient β_t for t equals June 2009 to zero. Thus, the vector β_t captures how much our outcomes of interest vary between period t and June 2009 in eligible and ineligible industries. We also add fixed effects of aggregate sector x time, represented in the equation by $\gamma_{s(c)t}$. Aggregate sectors are grouped by the first 2-digits of industry definition. This allows us to control for aggregate trends that are time-varying and possibly correlated with our treatment.

To summarize the effects across different phases of the program, we also estimate a pooled specification that interacts treatment status with indicators for the three program periods:

$$Y_{ct} = \theta_c + \gamma_{s(c)t} + \beta_1 \cdot P1_t \cdot D_c + \beta_2 \cdot P2_t \cdot D_c + \beta_3 \cdot P3_t \cdot D_c + \epsilon_{ct} \tag{2}$$

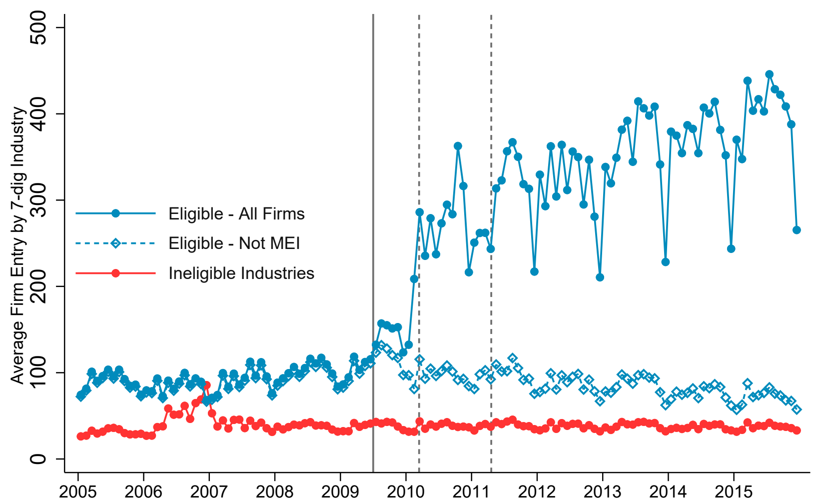
where $P1_t$ is a dummy that takes value one if the month is in the implementation phase between July 2009 and January 2010. $P2_t$ indicates that the month is between February 2010 and March 2011. And $P3_t$ indicates the period from April 2011 until December 2015. Thus the coefficients $\beta_t \quad t \in \{1, 2, 3\}$ reflect how much the outcomes differ between eligible and ineligible industries in periods 1, 2, and 3 relative to the pre-MEI program period.

Our identification assumption is that, in the absence of the MEI program, treated and control industries would have followed parallel trends. As discussed in Section 2, eligible and ineligible industries differ in composition, so we interpret these estimates as the average treatment effect on the treated (ATT). Each coefficient β_t reflects the average effect of the program on treated industries in period t, relative to July 2009.

Although the parallel trends assumption cannot be tested directly, we provide evidence that supports its plausibility. Figure 1 shows the average number of firm registrations in eligible and ineligible industries from 2005 to 2015. First, registration trends are similar prior to the program's implementation. Second, the post-program increase in registration is driven entirely by firms entering through the MEI tax regime. If unobserved shocks were driving the result, we would expect to see increases among non-MEI firms as well. Third, registration trends in ineligible industries remain flat, suggesting that firms are not switching into eligible industries to benefit from the program.[17]

[17] Figure A3 shows similar patterns using log-transformed outcomes, consistent with our main specifications.

Figure 1: Average Firm Registration By Industry Eligibility



This figure plots the average number of firms registering each month in eligible and ineligible industries. Eligibility is defined as the 7-digit CNAE level. The hollow diamond series shows the registration of firms not in the MEI program in eligible industries. Vertical lines represent the start of the phases of the program.

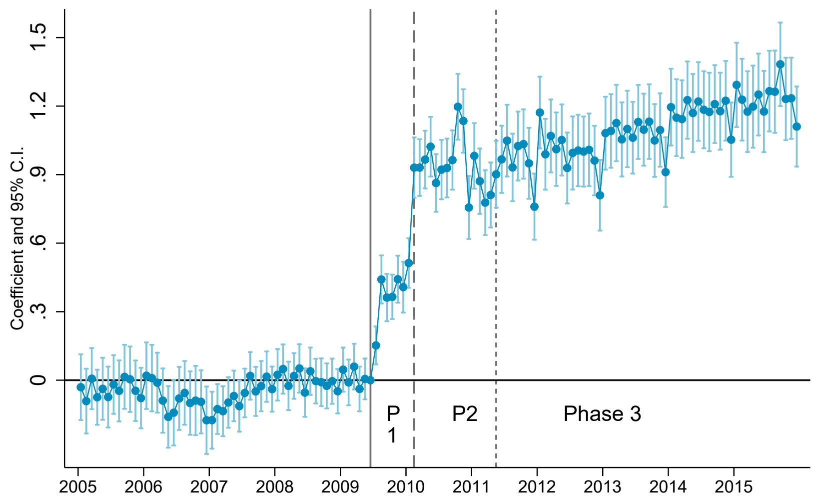
3.2 Estimates on Formal Firm's Entry and Number of Formal Firms

We begin by presenting our estimates of the MEI program's effect on formal firm entry, measured as the logarithm of firm registrations. One advantage of focusing on firm registration is that it is a flow measure and thus likely to respond more immediately to institutional changes than stock measures such as the total number of active firms. Combined with monthly data, this allows us to trace the program's impact with high temporal precision and examine how effects vary across implementation phases.

We find a large and sustained increase in firm entry in eligible industries. Figure 2 presents estimates of β_t from equation 1.

Three main results emerge from Figure 2. First, there is no evidence of differential pre-trends between treated and control industries, supporting the validity of the identification assumption. Second, program effects begin immediately in July 2009 and increase during Phase 2, when the program was fully operational. The modest effects in Phase 1 are expected, as not all states were yet authorized to register MEI firms and the integrated website had not been launched. Third, we find no substantial change in registration behavior between Phases 2 and 3. If anything, additional growth appears only in 2014–2015, rather than immediately after the 2011 tax cut. We discuss the implications of these patterns for understanding program mechanisms below.

Figure 2: Difference-in-Differences Coefficients on Firm Entry



This figure plots the vector of coefficients β_t and 95% confidence intervals from equation 1. The sample consists of a monthly panel at the 7-digit CNAE industry level from 2005 to 2015. The outcome is the logarithm of the number of firm registrations. Vertical lines mark the beginning of each program phase. Standard errors are clustered at the 7-digit CNAE level.

Table 2 presents estimates from the pooled specification in equation 2, summarizing the program's effects across phases. In our preferred specification—including fixed effects for 2-digit industries interacted with month—firm registrations increased by 166% in Phase 2 and 200% in Phase 3. Columns (3) and (4) report analogous results using the number of firms (in levels) as the outcome, yielding similar estimates.

Table 2: Estimates of Program's Effect on Firm Entry

	(1)	(2)	(3)	(4)
	Log(Firm Entry)		Firms Entry	
Eligible x Phase 1	.45***	.42***	60***	37***
	(.03)	(.036)	(9.3)	(7)
Eligible x Phase 2	1.1***	.98***	186***	122***
	(.045)	(.06)	(29)	(20)
Eligible x Phase 3	1.2***	1.1***	267***	191***
	(.055)	(.077)	(44)	(32)
Dep. Var. Mean of Treated	3.04	3.04	93.98	94.38
Month x 2 digit Ind. F.E.	No	Yes	No	Yes
Observations	143433	142404	143433	142404

This table presents estimates of coefficients $\{\beta_1, \beta_2, \beta_3\}$ from equation 2. Columns (1) and (2) use the logarithm of firm registrations as the outcome. Columns (3) and (4) use the number of firms (in levels). Standard errors are clustered at the 7-digit CNAE industry level.

Estimates on Number of Active Formal Firms: Next, we show the effects of the program on formal firm activity. We define a firm as active at period t if its date of entry was before t and its date of closure (if closed) was after t. If the firm had not closed until December 2015, it is defined as active from the moment it was registered until the end of our analysis period.

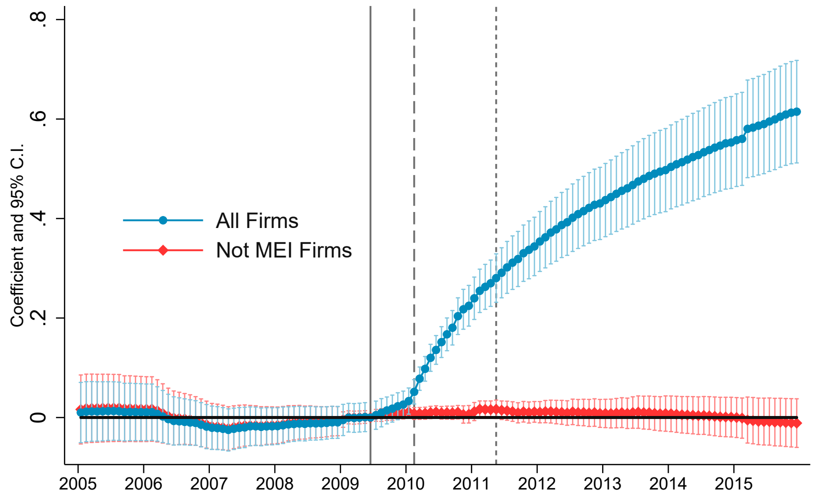
Our results are presented in Figure 3. It plots the estimates of model 1 for the log of the total number of active firms and the log of the number of active firms not in the MEI program. Our estimates indicate that the number of active firms in the eligible industries increased by 40% by December 2012[18] and by more than 86% by December 2015[19]. We also see that all the effect in the number of active firms is driven by MEI firms as the number of active firms not in the MEI program is stable through the whole period[20].

[18] Rocha et al. (2018) using RAIS finds an increase of only 4% by December 2012. As explained before, MEI firm owners are not obliged to report to RAIS, which makes their results largely underestimated.

[19] This is obtained by $exp(0.62) - 1 = 0.86$

[20] We also run the same specification in equation 1 for the log of firms closing. We show the results in Figure A5. We see a large increase in firm exit in eligible industries after the start of the program, and this is driven fully by MEI firms. But coefficients are smaller than the ones for firm registration, which leads to the positive result on formal firm activity.

Figure 3: Difference in Differences Coefficients on Active Firms



This figure plots the vector of coefficients β_t from equation 1 on the natural logarithm of the number of active firms in each 7-digit CNAE industry. The blue line shows the coefficients and 95% Confidence Interval of the regression using all firms as outcome whereas the green line shows the same statistics of a regression using only the number of Not MEI firms as outcomes. Vertical lines represent the start of the different phases of the program.

3.3 Mechanisms: Taxes or Entry costs?

The comparison between program phases provides insight into the mechanisms driving formalization. To disentangle these mechanisms, we exploit the staggered rollout of MEI registration across states. During Phase 1, only a subset of states was authorized to register MEI firms. The largest states—São Paulo, Rio de Janeiro, Minas Gerais, and the Federal District—joined in July 2009. Southern states and Ceará followed in September 2009. Remaining states became eligible at the start of Phase 2 in February 2010. We aggregate the data to the state × CNAE × month level and estimate the following regression:

$$Y_{uct} = \theta_c + \gamma_{us(c)t} + \beta_1 \cdot P1_t \cdot D_c + \beta_2 \cdot P2_t \cdot D_c + \beta_3 \cdot P3_t \cdot D_c + \beta_4 \cdot P4_t \cdot D_c + \beta_5 \cdot P5_t \cdot D_c + \epsilon_{uct} \quad (3)$$

where D_c indicates whether CNAE c was eligible for the MEI program. Fixed effects θ_c control for industry-level differences, and $\gamma_{us(c)t}$ are fixed effects for state × 2-digit sector × month. The five period indicators are defined as follows: $P1_t$ spans July–August 2009; $P2_t$ covers September 2009–February 2010; $P3_t$ spans March 2010–April 2011; $P4_t$ covers May 2011–December 2013; and $P5_t$ includes January 2014–December 2015. We divide Phase 3 into two subperiods to better isolate the effects of the 2011 tax cut.

Table 3 presents estimates of equation 3. Each column reports results for a group of states

Table 3: Estimates of Program's Effect on Firm Entry by State and Phases

	(1)	(2)	(3)
		Log(Firm Entry)	
	SP,RJ,MG,DF	PR,RS,SC,CE	Remainder
Eligible x July and August 2009	.37***	-.085***	-.082***
	(.042)	(.03)	(.029)
Eligible x [Sept.2009, Jan2010]	.39***	.27***	-.09***
	(.039)	(.039)	(.026)
Eligible x [Feb.2010, Apr.2011]	.71***	.45***	.44***
	(.054)	(.049)	(.055)
Eligible x [May.2011, Dec.2013]	.66***	.46***	.37***
	(.054)	(.047)	(.047)
Eligible x [Jan.2014, Dec.2015]	.77***	.56***	.44***
	(.062)	(.054)	(.054)
Dep. Var. Mean of Treated	1.66	1.27	.84
State x Month x 2 digit Ind. F.E.	Yes	Yes	Yes
Observations	308388	322559	619446

This table presents estimates of coefficients $\{\beta_1, \beta_2, \beta_3, \beta_4, \beta_5\}$ from equation. The outcome is the log of firm registration at a given month for a given 7-digit industry. Standard Errors are clustered at the industry level. The groups of state in the header represent the sample used in that respective column.

defined by the timing of MEI program rollout.[21]

The estimates suggest that reductions in fixed registration costs played a central role in driving formalization. For the two earliest adopter state groups, the only major change from Phase 2 onward was the introduction of the streamlined online registration system. Comparing coefficients across phases, we find that firm entry nearly doubled after the reduction in registration costs.

These findings also contradict the interpretation that the tax cut from 11% to 5% of the minimum wage in May 2011 was the primary driver of formalization. Rocha et al. (2018), using limited data, find effects only after this tax cut and attribute the program's impact to lower tax burdens. In contrast, we find comparable increases in firm entry across all state groups in both Phase 2 and the early part of Phase 3, indicating that the original 11% tax was not a prohibitive barrier for most microentrepreneurs.

While our results emphasize the importance of entry costs, we do not attribute all effects solely to their reduction. First, we observe a measurable increase in firm registrations even during the early months of Phase 1, when the website was not yet operational. Second, as

[21] Figures A6 and A7 show monthly coefficients using log and level outcomes, respectively.

noted in Section 2, the 11% flat tax likely represented a reduction for a substantial share of microentrepreneurs. Third, the MEI regime simplified compliance by eliminating bookkeeping requirements and allowing a single monthly tax payment. Nonetheless, the sharp increase in registrations following the introduction of the online system strongly suggests that fixed entry costs were a key constraint for potential formal firms.

3.4 Additional Results and Robustness Checks

One potential concern is that the low cost of MEI registration may have induced microentrepreneurs to register without ever becoming active, thereby overstating the program's impact on formal firm activity. If many registrants quickly ceased operations or never began them, the observed increase in the number of active firms would exaggerate the true effect. To address this, we draw on data from the *Perfil do MEI* survey, a qualitative study conducted annually by SEBRAE, a federal agency supporting small businesses.[22] From 2012 to 2015, approximately 90% of respondents reported being currently active. In 2015, the survey further broke down the inactive group, revealing that one-third of them intended to start operations soon. These figures suggest that the CNPJ registry provides a reliable measure of formal activity and support the validity of our estimates. Summary statistics from the survey are reported in Table A3.

As an additional robustness check, we re-estimate our baseline difference-in-differences models using a matched sample of control industries. Specifically, we implement a one-to-one propensity score matching procedure without replacement, using 2008 industry-level characteristics: the number of firm entries, the number of active establishments, and the share of firms with limited liability. This procedure yields a balanced sample of treated and control industries. Table A2 and Figure A8 show that the estimated effects remain very similar to those from our main specification using the full control sample.

4 Effects on Formalization and Microentrepreneurs' Earnings

In this section, we use household survey data and combine RAIS and CNPJ administrative records to estimate the effects of the MEI program on informality and the earnings of microentrepreneurs. We also present descriptive evidence on firm size and rule out alternative explanations for our findings. We show that the program formalized a large share of microentrepreneurs but had no measurable effect on their earnings.

[22] SEBRAE is an autonomous entity linked to the Brazilian federal government that provides assistance and research on small firms and microentrepreneurs. From 2011 to 2015, SEBRAE conducted the *Perfil do MEI* survey through randomized phone interviews with MEI registrants, collecting detailed information on business activity.

4.1 Descriptive Statistics

The share of formal microentrepreneurs increased by 15 percentage points in the six years following the program's implementation[23]. This represents a 79% increase in the formality rate. Figure 4a shows that the share of formal microentrepreneurs was stable at around 18–19% from 2002 to 2008, before rising sharply after MEI was introduced.

The composition of microentrepreneurs is not driving these changes. This is particularly relevant in the period between 2002 and 2015, when Brazil observed a demographical change, with a large increase in education levels.[24] To investigate the effects of composition changes on formalization rates of microentrepreneurs, we residualized the probability of an individual being formal on a series of observable characteristics.[25] We observe that the pattern controlling for observable characteristics is very similar to the unadjusted data in Figure 4a. There are no changes between 2002 and 2008, and after MEI's implementation, there is a large increase in the share of formal microentrepreneurs.

As an additional piece of evidence, in the red series in Figure 4b, we replicate the same exercise but with a sample of employees. The 2000's decade was a period of formalization of employees (Haanwinckel and Soares, 2021). However, the increase in microentrepreneurs' formality does not appear to be concurrent with the trend of employee formality. We see this as additional evidence of the MEI program as a large driver of the formalization of microentrepreneurs.

Heterogeneity: We decompose our sample in different ways to better understand how changes in informality rates differ across groups. We show these results in Figure A10, where each Panel shows a different sub-sample comparison.[26] We find that women face a larger change in their informality rate than men. By 2015 female microentrepreneurs are more likely to be formal than male ones. One possible reason for why this is happening is because the MEI program grants paid maternity leave to women who contributed to the tax authority for at least 10 months. When looking at race, white individuals have a slightly larger increase in absolute terms than nonwhite ones, although the percentage increase is higher for the nonwhite population, which starts from a significantly lower baseline. We also see that after the implementation of the MEI program, there is a large increase in the formality rate of microentrepreneurs with high school education and those in the middle of the earnings distribution.

[23] In what follows, we exclude entrepreneurs in agriculture. The same patterns including agriculture are shown in Appendix Figures A9 and A11. In Appendix B, we also replicate aggregate trends using PME data, as in Rocha et al. (2018).

[24] Haanwinckel and Soares (2021) argues that these changes in education levels promoted a large share of the formalization of Brazilian employees during that time.

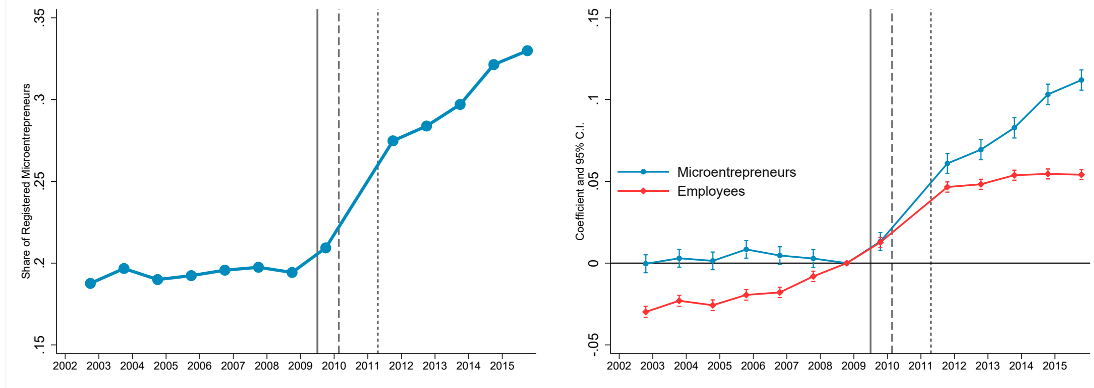
[25] We estimate through OLS the following regression with the sample of all microentrepreneurs between 2002 and 2015:

$$\text{Formal}_{it} = \alpha_t + \Gamma X_{it} + \epsilon_{it} \tag{4}$$

where α_t are time dummies(with t=2008 was normalized to zero), and X_{it} is a vector of covariates that include gender, race, the natural logarithm of monthly income, a second order polynomial on age and fixed effects on the years of schooling, industry, and state. We plot our estimates of α_t in the blue series in Figure 4b.

[26] In addition to the trends by different groups, in Appendix C we provide regression based analysis of the characteristics of compliers which give us similar insights to our figures.

Figure 4: Share of Formal Microentrepreneurs and Employees



(a) Microentrepreneurs **(b)** Controlling by Observables

Microentrepreneurs are defined as individuals who declare themselves as self-employed or employers who hire at most one employee. We define a formal microentrepreneur as those who declare to be contributing to social security through their job. Panel A plots the share of microentrepeneurs between 18 and 60 years that are formal by year. Panel B plots the coefficients α_t of two separate regressions. The blue one, is with the sample of microentrepreneurs. The red one with the sample of individuals working as employees of a firm. All results were built with PNAD 2002 - 2015.

4.2 Empirical Strategy

Next, we describe our empirical strategy to evaluate the effects of the program on the formalization of microentrepreneurs and to evaluate if the program had consequences on their earnings. We extend our industry-level analysis to PNAD data, with appropriate adjustments to accommodate its structure.

We begin by organizing PNAD data into an industry x year panel. We calculate the total number of microentrepreneurs, their average earnings in the job, and the number of formal microentrepreneurs. We then use the crosswalk from IBGE between fiscal CNAEs in which the MEI program eligibility was defined, and the *CNAE domiciliar*, the industry classification used in PNAD. With our panel organized, we estimate the following regression:

$$Y_{ct} = \theta_c + \gamma_t + \sum_{\substack{t=2002 \\ t\neq 2008}}^{2015} \beta_t \cdot D_{ct} + \Gamma X_{ct} + \epsilon_{ct} \tag{5}$$

where θ_c are *CNAE domiciliar* fixed effects, γ_t are year fixed effects, and D_{ct} is a indicator that takes value one for eligible CNAEs in period t. The main difference relative to equation 1 is that we are not able to include 2-digit by time fixed effects in this case due to the less-granular PNAD data. We are also able to include time varying controls

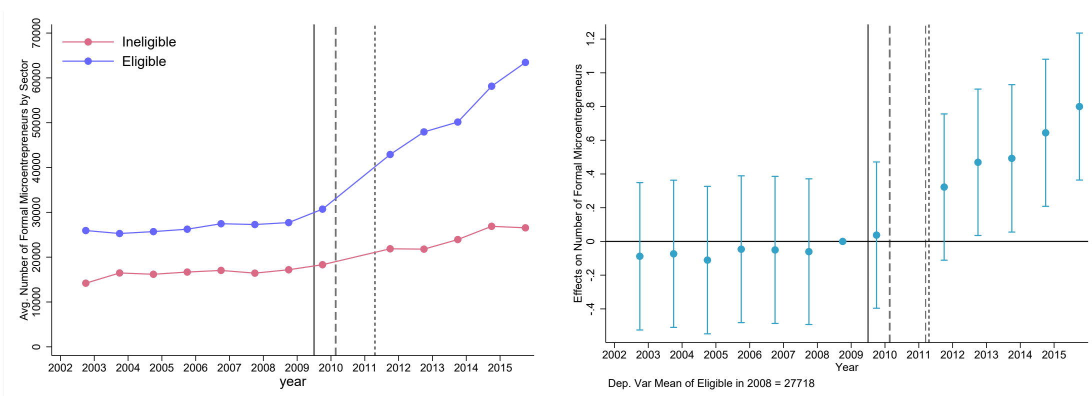
The identification assumption in this model is the same as before: in the absence of the MEI program, eligible and ineligible industries would follow parallel trends. As before, eligible and ineligible industries are not the same, thus our coefficient is interpreted as an Average Treatment Effect on Treated industries.

4.3 Results

We begin by showing that formalization substantially increased in eligible industries relative to ineligible ones. Those results are shown in Figure 5. In Panel (a) we plot the average number of formal microentrepreneurs in eligible and ineligible industries. We observe that both groups follow similar trends prior to 2009, despite an initially higher number for eligible industries. However, after the reform, we observe a substantial increase in the number of formal microentrepreneurs in eligible industries, whereas we do not see the same patter in ineligible ones.

This increase translates into our estimates of treatment effects, which we present in Panel (b) of Figure 5. To facilitate the interpretation, we divide our estimates by the baseline number of formal microentrepreneurs in treated industries in 2008.[27] Our estimates indicate that MEI caused the number of formal microentrepreneurs in eligible industries to increase by almost 80% by 2015.[28]

Figure 5: Effects on Formalization of Microentrepreneurs by Industry Eligibility



(a) Formal Microentrepreneurs - Trends

(b) Formal Microentrepreneurs

This figure shows formalization of microentrepreneurs across eligible and ineligible industries using PNAD data. Panel (a) shows the number of formal microentrepreneurs among eligible and ineligible industries. Panel (b) plots the coefficients β_t from equation 5 divided by the average of the outcome variable in 2008. The sample comprises all microentrepreneurs in PNAD between 18 and 60 years old. We include as controls the share of women, the share of nonwhite, average years of schooling, average hours worked, and share of microentrepreneurs in each state.

Next, we show that the MEI tax system did not have effects on microentrepreneurs' earnings in eligible industries. Our results are presented in Figure 6. In Figure 6a we again show labor earnings in an average eligible and an average ineligible industry. There is no clear change in

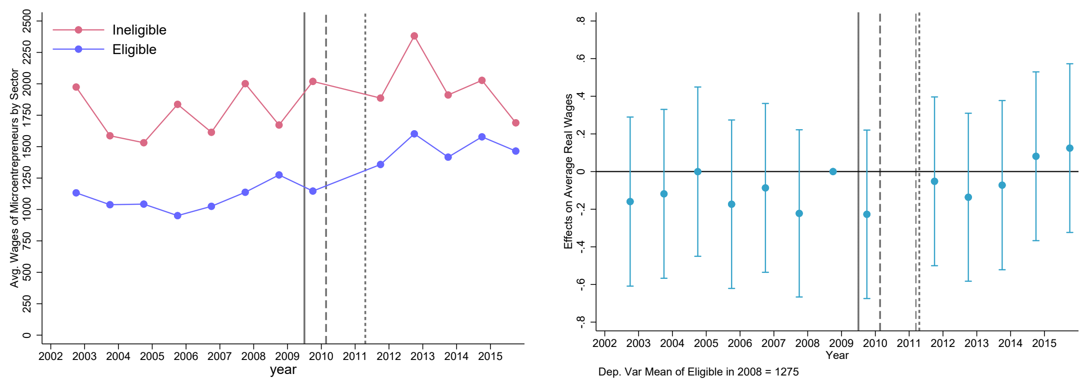
[27] We present our coefficients without dividing by the baseline in Figure A12.

[28] In Figure A14 we show similar results using the number of microentrepreneurs in that given industry as the outcome. We observe no change, indicating that increases in microentrepreneurship in eligible industries are not driving our findings. We also find no effects on the number of employees from firms in these sectors using PNAD data. This is presented in Figure A15.

average labor earnings between the two sets of industries.

We report our estimated treatment effects in Figure 6b. None of the coefficients is statistically different from zero. Although the estimates are not very precise, the post-policy point estimates for β_t hover around zero.

Figure 6: Effects on Labor Earnings of Microentrepreneurs



(a) Average Labor Earnings - Trends **(b)** Average Labor Earnings

This figure shows the average labor earnings of microentrepreneurs across eligible and ineligible industries using PNAD data. Labor Earnings are adjusted to December 2018 values. Panel (a) shows the number of formal microentrepreneurs among eligible and ineligible industries. Panel (b) plots the coefficients β_t from equation 5 divided by the average of the outcome variable in 2008. The sample comprises all microentrepreneurs in PNAD between 18 and 60 years old. We include as controls the share of women, the share of nonwhite, average years of schooling, average hours worked, and share of microentrepreneurs in each state.

Our findings on the effects of formalization on microentrepreneurs' earnings are consistent with the literature. In a recent review of the literature, Ulyssea (2020) argues that despite some mixed results, most estimates of the effects of formalization find no statistically significant impact on various measures of firm performance, including sales, profits, and employment. One exception is De Mel et al. (2013), who finds significant effects, driven by a small share of treated firms experiencing substantial growth. In what follows, we provide descriptive evidence on the universe of MEI firms, in which we can look at detailed measures across the distribution of firm size.

4.4 Descriptive Statistics on Firm Size Using Adminstrative Data

We complement the household survey analysis with descriptive statistics on firm size covering the universe of MEI firms. We combine the CNPJ data with administrative records that comprise the near universe of firm ownership in the country to create a data set that follows firm growth at the entrepreneur level[29]. This data set allows us to observe if individuals who register their firms through the MEI program grow after registration in terms of the number of workers employed and the wage bill.

[29]The procedures to build the data set are carefully described in Appendix A.1

We find that most microentrepreneurs who register as MEI firms do not grow according to observable measures. In Panels (A) and (B) of Figure 7 we plot the distribution of firm size of firms from microentrepreneurs that registered as MEI. In panel A we show the firm size distribution conditional on being active at t years after you registered. Two years after registering as MEI, only 3% of microentrepreneurs are hiring at least one worker. Four years after the firm's registration, only the 99th percentile of active microentrepreneurs hire more than one worker (2 workers). In panel B of Figure 7 we show the firm size distribution conditional on having at least one formal employee. Four years after registering as MEI, the 99th percentile of individuals hiring at least one formal employee only has 14 employees. Furthermore, even among those microentrepreneurs that hire someone, the vast majority of them are still hiring only one employee.[30]

To benchmark the distribution of workers hired by these microentrepreneurs, in Panels (C) and (D) of Figure 7 we show them together with the firm size distribution in the Brazilian economy in 2015. The distribution for the whole economy was calculated only using the private sector for-profit firms that hire at least one worker (excluding public sector and publicly administrated firms, which are the largest employers in the country, and non-profit institutions). We see the stark contrast between firms' firm size distribution from microentrepreneurs registered as MEI and the firm size distribution in the economy. [31]

The evidence from our longitudinal analysis shows that microentrepreneurs who register their firms as MEI firms are not growing to become successful entrepreneurs. The observed (lack of) growth of these microentrepreneurs is enough for us to discard large gains of formalization for this population. We see that the bulk of them remain self-employed, and those that become employers remain small.

4.5 Additional Results

In what follows, we show additional results using both PNAD and administrative data to rule out competing narratives on the effects of the MEI program.

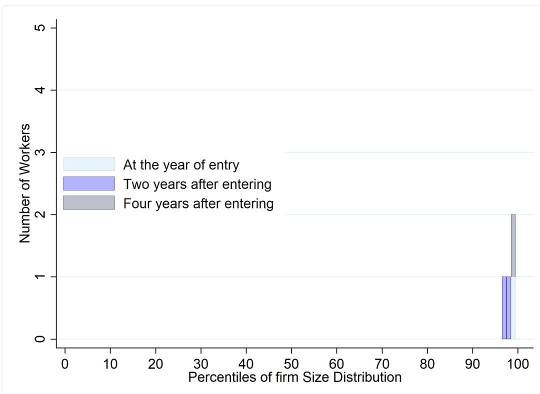
Occupational Change: An alternative explanation for the effects of the program on formal firm entry and the number of active firms might be the transition of individuals from being out of the labor force into formal microentrepreneurship or the shift of workers from formal or informal employment into formal microentrepreneurship. We next show evidence that these two explanations are not driving our findings.

We observe no significant changes in the share of the adult population working around the implementation of the program. This is shown in Figure A16a, where we use PNAD data to calculate the share of individuals between 18-60 that report to be working per year. We also

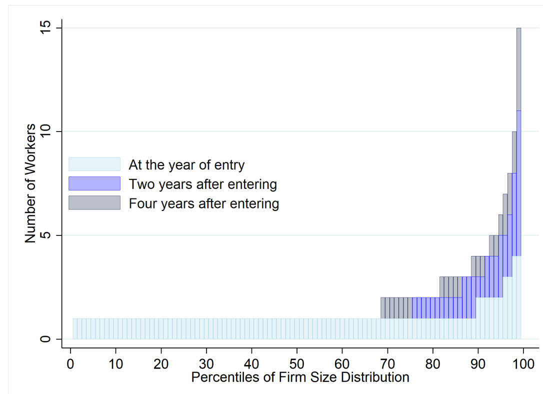
[30] The firm size distribution is fully detailed in Table A4.

[31] We see similar results when looking at firms' wage bills in Figure A18. The wage bill is defined as the total amount of wages the firm paid in December of the respective year. These results are useful because they discard a possible concern that these firms are still small but highly productive. If that were the case, we would expect that the employees there would have better wages.

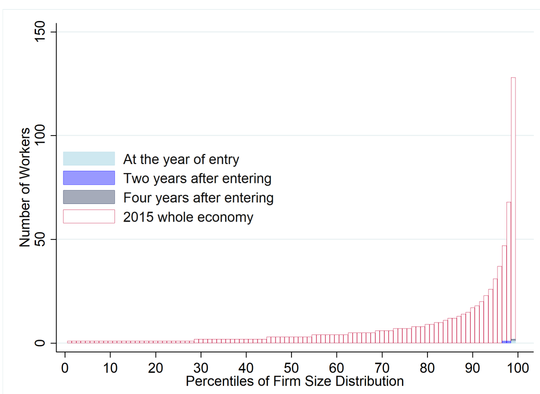
Figure 7: Firm Size Distribution of Microentrepreneurs that Register as MEI



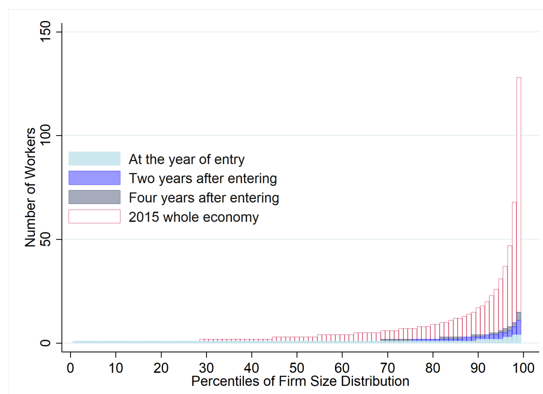
A. Wage Bill Unconditional



B. Firm Size - Conditional



C. Unconditional with RAIS



D. Conditional With RAIS

Notes: This figure is built from the panel of MEI firms combined with ownership data and the economy distribution of firm size from RAIS. Firm size is calculated as the total number of individuals that were reported in RAIS as formally employed by December 31st at the respective firm. On panel (A), we input zero workers to an active MEI firm that does not appear in RAIS. On panel (B) we restrict our sample to microentrepreneurs that do appear in RAIS. Panels (C) and (D) are the same as in (A) and (B) but adding the observed economy Firm Size distribution in RAIS in 2015.

observe no changes in the type of occupation in which individuals are employed. This can be seen in Figure A16b, where we divide the working population in 4 main groups. The household survey data indicates that there are no large changes in the composition of the workforce around the implementation of the program.

These patterns indicate that individuals are not becoming microentrepreneurs because of the program. Thus ruling out occupational change as a driver of our findings in firm entry or firm activity.

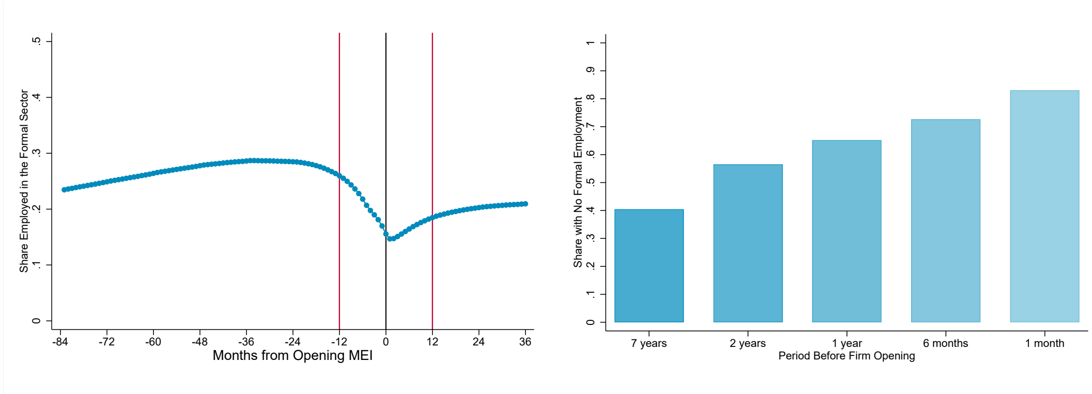
Transitions Between Formal Employment and Formal Microentrepreneurship: One concern frequently discussed in the Brazilian Policy debate is that workers leave formal employment ties to become MEI firm owners because of its tax benefits while still doing the same job for the firm that used to hire them through an employment tie (Alvarez, 2024). This behavior could potentially not show up in the occupation shares of PNAD because the individual might still declare herself an employee, although her formal link is through a MEI firm. If all formal microentrepreneurship increase was driven by individuals that would have been formal through employment, then the MEI program would not have generated a reduction in informality, just a reallocation of formal workers to formal microentrepreneurship

To investigate this possibility, we track transitions between formal employment and formal microentrepreneurship. Our data allows us to individually match microentrepreneurs to their formal employment before and after registering their firm.[32] We build a monthly panel that tracks formal labor market information of all microentrepreneurs for whom we have Social Security Numbers. In each period, we observe if the individual was employed in the formal sector and was a registered microentrepreneur.

We see in Figure 8a that transitions from formal employment to formal microentrepreneurship are smooth around the period of registration of the firm. To construct it, we pooled all MEI owners for whom we have the Social Security Number and tracked if they worked as formal employees or not up to 7 years before and three years after they registered as MEI firm owners. Around one year before the firm's registration, the share of individuals that are employed starts to decrease from a relatively stable level of 28%. We see that 15% of individuals register as MEI firm owners while holding a formal employment tie. This decrease shows transitions from form, but it also shows that the large majority of MEI firm owners are not transitioning from employment to microentrepeneurship, rather going from an "informal" state where we do not observe them in administrative records, to formalization through the MEI program. Figure 8a also indicates that individuals gaming the labor tax system is not driving the registration of workers as MEI. If individuals were avoiding labor taxes to become MEI, we would expect a sharp decrease around the period they register as MEI, indicating that they were dropping their

[32] This is due to a simple coincidence. Starting in Feb. 2010, MEI firms are named using the combination Owner's Name + Social Security Number. We exclude from this analysis the small share of individuals who have more than one MEI firm. We then match their social security number to RAIS, the Brazilian matched employer-employee data that covers the universe of the formal labor market.

Figure 8: Share of Microentrepreneurs that were employed in the formal Sector



(a) Share Working **(b)** % No Formal Work Before Opening MEI

Panel A plots the share of individuals that were employed in the formal sector relative to the month at which they opened their MEI. Panel B plots the share of individuals that had no formal labor market tie in the periods before opening their firms. The sample comprises all microentrepreneurs with a valid Social Security Number in CNPJ that registered between Feb. 2010 and Dec. 2015. Being employed in the formal sector means having a valid contract registered at RAIS.

formal employment tie to register a firm as MEI.[33]

In Figure 8b, we show the share of microentrepreneurs that had no formal tie in the period before opening their firm. We find that 40% of the MEI firm owners did not have a single formal labor market contract in the 7 years before opening their firms. This number is increasing when we get closer to the MEI opening. For 70% of microentrepreneurs, their MEI firm was their only first formal labor market tie in a span of a year.

5 Fiscal Consequences - Revenue Gains and Costs with Social Security

In this final section, we examine the fiscal consequences of the program. Informality is often linked to limited tax collection capacity in developing countries. It is therefore essential to assess whether a program that reduces taxes to promote formalization can generate net fiscal benefits.

Understanding this trade-off requires analyzing the costs of formalization. As shown, commonly cited costs—such as firms opting for the simplified tax regime over the regular one—do not materialize in our context. However, a significant fiscal cost arises from the social security benefits newly formalized individuals become eligible for.

We begin by documenting the revenue gains under the MEI tax system, then estimate the program's effects on social security benefits. Finally, we calculate the overall fiscal impact on the government and conclude with a discussion of the broader implications.

[33] In Figure A17 we present the same exercise but dividing the sample by the year of entry of the MEI firm. This allows us to see the labor market behavior for more years after the MEI firm was opened. We find that MEI firm owners become slightly more likely to transition from the formal labor market to formal microentrepreneurship in more recent cohorts. The behavior after registering as MEI is similar across cohorts.

Revenue Gains: We quantify the increases in government revenue caused by the program, one of the commonly advocated reasons to reduce informality. In the previous sections, we showed that the program drove microentrepreneurs to the formal sector, implying a higher number of formal firms. At the same time, to induce formalization, the program substantially reduced tax rates. Furthermore, there is not necessarily full compliance with paying taxes, which can further reduce the gains in revenue for the government and generate challenges in observing the effects of the program on tax revenues.

To overcome this, we use data from the *Anuário Estatístico da Previdência Social* (AEPS).[34] Yearly, the *Instituto Nacional de Seguridade Social* (INSS), the national autarchy responsible for social security in Brazil, reports its revenues and the number of contributors to the MEI program. This allows us to observe the total value contributed by individuals in the program and also to estimate their compliance with contributions. Given that we find no substitution patterns of individuals shifting to the MEI program from other formal activities, these numbers provide estimates of the causal effect of the program on revenues. We summarize these statistics in Table 4.

The MEI program did not represent significant revenue gains relative to the Brazilian government collection. By 2015, the MEI program represented a yearly revenue of almost 800 Million Reais to the government. This is around 0.1% of government revenues and 0.3% of total social security contributions.[35]

We also find that a large share of registered microentrepreneurs do not comply with their contributions. In the period between 2010 and 2015, the share of defaulters lie around 40-45% of the full sample.[36] Low compliance, however, does not explain why the MEI program does not have meaningful impacts on government revenues, as a simple calculation indicates that even under full compliance, its revenues would represent a very small share of both total Government and Social Security collection.

Social Security Coverage: Next, we quantify the effects of the MEI program on social security coverage. As in many countries (Levy, 2008), access to social security in Brazil is largely conditional on formal workforce participation. Thus, by driving microentrepreneurs out of informality, the MEI program also started covering them in the social security system.

We use data that covers the universe of social security benefits paid by the INSS. This data is organized at the benefit level, and allows us to track the dates in which they were issued and ended, their location and the *class* of coverage that generated the benefit.[37]

Our analysis consists of a difference in differences model that compares how benefits evolved

[34] See Appendix A.1 for a description of the AEPS data.

[35] Total government revenue in 2015 was 820 Billion Reais and Social Security Contributions amounted to 240 Billion Reais (in 2009 Values).

[36] These numbers are consistent to the findings from (Bosch Mossi et al., 2015), who estimate an average default rate of 41.6% for MEI affiliates between 2011-2015 using similar administrative data from the Ministry of Social Security.

[37] See Appendix A.1 for a detailed description of the INSS benefits data.

Table 4: Descriptive statistics of the MEI contributions to social security

Year	Number of MEI Firms	Number of MEI Contributors	$ MEI Contributions (BRL 2009)	Percentage of MEI Defaulters
2010	949,720	557,540	167,402,668	41%
2011	1,844,149	995,289	243,531,412	46%
2012	2,854,975	1,523,131	335,067,859	47%
2013	3,884,171	2,164,620	522,061,304	44%
2014	4,928,719	2,854,889	686,130,839	42%
2015	6,019,331	3,395,337	795,306,794	44%

Notes: This table presents descriptive statistics of the MEI contributions to the Brazilian pension system. The first column following the year brings counts of the CNPJ data as a reference. The remaining information comes from the *Anuário Estatístico da Previdência Social*/INSS.

before and after the implementation of the MEI program for different classes of coverage. We consider the Individual Contributors class to be our treated group. That encompasses entrepreneurs and self-employed individuals. We use all the remaining affiliate types in the control group. That includes wage-dependent workers, special insured, and facultative affiliates.[38]

We then aggregate our data at a quarter x state x class level and estimate the following equation:

$$Y_{out} = \alpha_o + \delta_{ut} + \sum_{t=2005}^{t=2015} \beta_t \cdot D_{ot} + \varepsilon_{out} \tag{6}$$

where α_o, δ_{ut} are fixed effects by State times quarter, and α_o is an indicator for the class. D_{ot} is an indicator that the class is treated (Individual Contributors), and the period is equal to t.

We focus our analysis on two primary outcomes: the log of benefits issued and the log of total benefits that are active at a given moment. The identification assumption is that in the absence of the program, the trends of our outcomes would be parallel across different classes of individuals covered by social security.

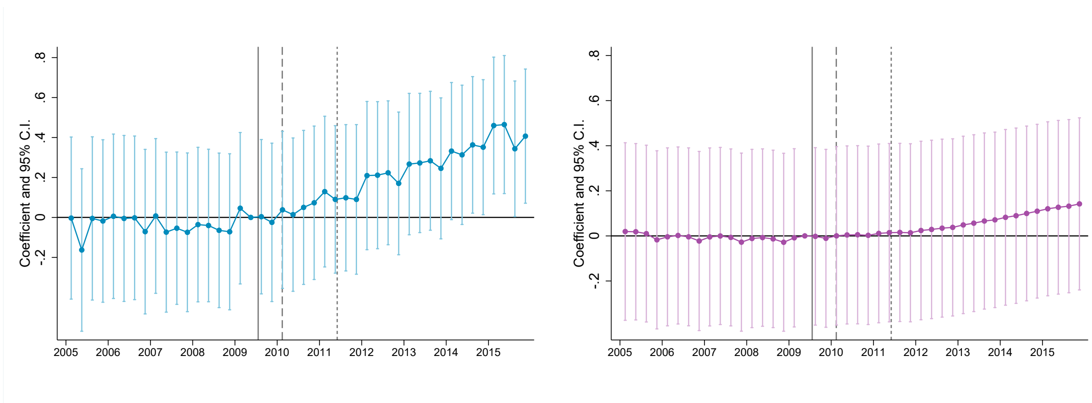
We estimate that the MEI program substantially increased the number of social security benefits issued to individual contributors by the end of 2015. Our point estimates also indicate an increase in the number of active benefits, although they are not statistically different from zero. Our estimates are shown in Figures 9a and 9b.[39] We highlight two important patterns in our results that corroborate to the causal interpretation of our findings. First, we see the absence of any differential trends in our coefficients before the program implementation, indicating the validity of our identification assumption. Second, the effects are delayed relative to the start of the program. This is reassuring as MEI firm owners have to contribute to Social Security for at least 10 months to be covered by the majority of benefits from INSS. This also lines up with

[38]Refer to appendix A.1 for a complete description of the INSS affiliate classes.

[39]In Figure A19, we estimate the same equation but aggregate the data by Month and Semester. We find similar patterns as our results at the quarterly level.

our findings on the number of MEI contributors, which is increasing over the period.

Figure 9: Effects on the Social Security Benefits



(a) Log of Benefits Issued **(b)** Log of Active Benefits

This Figures show estimates β_t from equation 6. Panel (a) uses as the outcome the log of benefits issued, whereas Panel (b) shows the log of benefits that are active in the given quarter. The benefit-level data is aggregated at the State x Quarter x Class level. Vertical lines indicate the start of the different phases of the program.

We summarize our findings by estimating two different equations that group different periods according to the phases of the program. These estimates are going to be used when analyzing the costs and benefits of the program. The equations are the following:

$$Y_{out} = \alpha_o + \delta_{ut} + \beta_1 \cdot D_o \cdot \text{Phase } 1_t + \beta_2 \cdot D_o \cdot \text{Phase } 2_t + \beta_3 \cdot D_o \cdot \text{Phase } 3_t + \varepsilon_{out} \tag{7}$$

$$\begin{aligned} Y_{out} = \alpha_o + \delta_{ut} + \beta_1 \cdot D_o \cdot \text{Phase } 1_t + \beta_2 \cdot D_o \cdot \text{Phase } 2_t + \\ \beta_3 \cdot D_o \cdot \text{Apr11-Dec13}_t + \beta_4 \cdot D_o \cdot \text{Jan14-Dec14}_t + \beta_5 \cdot D_o \cdot \text{Jan15-Dec15}_t + \varepsilon_{out} \end{aligned} \tag{8}$$

The only difference between them is that in equation 8, we separate the last phase of the program into three parts. This allows us to capture the dynamic effects of the program on social security coverage.

We estimate that the MEI program increased by around 55% the number of social security benefits issued to individual contributors by the end of 2015. Our point estimates also indicate an increase of 15% in the number of active benefits. [40] We show our estimates of the coefficients from equations 7 and 8 in Table 5. When grouping different quarters together, we also see that the estimates on the number of active benefits become more precise, and we can reject that they are equal to zero.

[40] The difference between our findings on issued and active benefits can be explained by the effects of the MEI program being concentrated on shorter-term benefits. We replicate our exercise decomposing our outcome by type of benefit in Table A5. There, we see that longer-term benefits such as retirement are not affected by the program.

Table 5: Estimates of Program's Effect on Social Security Benefits

	(1) Issued	(2) Active	(3) Issued	(4) Active
Treated x Phase 1	.024	-.00054	.024	-.00054
	(.11)	(.11)	(.11)	(.11)
Treated x Phase 2	.096	.011	.096	.011
	(.072)	(.072)	(.072)	(.072)
Treated x Phase 3	.31***	.075		
	(.044)	(.046)		
Treated x (Apr11 - Dec13)			.23***	.043
			(.051)	(.054)
Treated x (Jan14 - Dec14)			.37***	.1
			(.064)	(.076)
Treated x (Jan15 - Dec15)			.45***	.14*
			(.064)	(.075)
Dep. Var. Mean of Treated	7.34	9.52	7.34	9.52
Observations	2376	2376	2376	2376

This table presents estimates of coefficients $\{\beta_1, \beta_2, \beta_3, \beta_4, \beta_5\}$ from equations 7 and 8. Columns (1) and (3) use as outcomes the log of benefits issued. Columns (2) and (4) show results using the log of active benefits as the outcome. The benefit-level data is aggregated at the State x Month x Class.

Costs of the program to the Government: With our findings, we can calculate what are the net government costs of the program. We calculate it as the changes in expenditures and the gains in revenue caused by it:

$$\text{Net Gov. Cost} = \Delta E - \Delta R$$

where ΔE denotes the government expenditures with the policy, and ΔR the gains in government revenue caused by it.

We use the revenue collected from contributions of MEI firm owners as our measure of gains in government revenue (ΔR). Given our findings that (i) the program did not shift firms from the regular tax systems to the MEI tax system, (ii) microentrepreneurs that registered as MEI would have been informal in the absence of the program, it is reasonable to assume that in the absence of the MEI program, the revenue from contributions would not have been collected. Thus, the changes in revenues caused by the policy are equivalent to what was collected by MEI contributors.

The main expenditures with the program come from social security benefits given to formalized individuals. Using our findings from the previous section, we can calculate how much this represented in terms of government funds. To do so, we estimate the predicted benefits handed to the treatment group in the absence and the presence of the policy in each state and each period, $\hat{B}^0_{ut}, \hat{B}^1_{ut}$ respectively.

$$\hat{B}^0_{ut} = \hat{\alpha}_o + \hat{\delta}_{ut} \tag{9}$$

$$\hat{B}^1_{ut} = \hat{\alpha}_o + \hat{\delta}_{ut} + \hat{\beta}_t \cdot D_{ot} \tag{10}$$

where $\{\hat{\alpha}_o, \hat{\delta}_{ut}, \hat{\beta}_t\}$ are the coefficients estimated in equation 6.

We then compute the exponential of $\hat{B}^0_{ut}, \hat{B}^1_{ut}$ and sum the total number of benefits across states in each period. This gives us estimates of the actual number of benefits in a given period and the counterfactual in the absence of the program. We plot these estimates together with the difference between them in Figure A20.

Our estimates indicate that in 2015, social security had, on average 322,280 benefits more to Individual Contributors. This implies 2.038 Billion BRL more in annual expenses for the government.[41]

Our findings thus imply that in 2015, the net government cost of the program was 1.243 Billion Brazilian reais:

$$\text{Net Gov. Cost in 2015 (Billions) } = \Delta E - \Delta C = 2.038 - 0.795 = 1.243$$

or around 207 Brazilian Reais per registered microentrepreneur per year.

In relative terms, our findings imply that the expenditures with social security are around 2.56 times higher than what the government collects as tax revenue.

Discussion: Our findings corroborate the idea that programs that substantially reduce taxes are not effective ways of increasing government revenue by bringing informal individuals to the formal sector (Ulyssea, 2020).

However, the simple government accounting exercise ignores the benefits of social security coverage to the microentrepreneur population. Besides the immediate utility generated to those who receive the benefits, there is a large supporting evidence of externality gains of social security coverage. In Brazil, for example, GC Britto et al. (2024) estimate that maternity benefits issued to unemployed women by social security can reduce the criminal activity of first-time fathers.

Furthermore, the government accounting exercise also does not take into account other po-

[41] We multiply the number of benefits by the monthly minimum wage of 527. (788 adjusted to 2009 BRL levels) and multiply by 12 months.

tential indirect benefits of higher registration levels in an economy. By keeping track of individuals' earnings, governments can potentially improve the targeting of public policies such as cash transfers. Another eventual benefit can come from third-party reporting of MEI activities. If larger firms are hiring the services of microentrepreneurs, they can now issue invoices and help them keep track of the balance sheets. These indirect benefits are not estimated in this paper but are important aspects of the policy that can reduce the revenue losses incurred by the MEI system.

Lastly, given that we found no differential effects on firm registration of the tax cut from 11% to 5% and no clear change in tax compliance between before and after the tax cut, revising the tax subsidy to the original rate could be a useful adjustment to the program, thus reducing the deficit generated by social security benefits.

6 Conclusion

To this day, the combination of high informality levels and a large population working as microentrepreneurs remain a common feature of most developing economies. Understanding the causes of high informality levels among microentrepreneurs and their consequences is thus a fundamental question for economic development.

In this paper, we studied the MEI program, a unique natural experiment that pushed microentrepreneurs to register as formal firms at the expense of lower taxes and entry costs. With unique data, we showed how large reductions in formalization costs made by the program substantially increased the number of formal firms, driven by a large increase in formal firm entry. We show that this is mainly driven by the simplification and reduction in registration costs. We also demonstrate that new formal firms were driven by microentrepreneurs who would have been informal in the absence of the program.

By reducing substantially the taxes for microentrepreneurs, the MEI program ends up being deficitary. We estimate that costs incurred in social security benefits given to newly formalized individuals are 42% higher than the total revenue collected from newly formalized firms. The program also did not foster large growth for newly formal microentrepreneurs.

Our findings can hopefully guide policy targeting microentrepreneurs' informality. We show that they are very responsive to simplified tax systems that generate substantial reductions in the costs of formalization. We also illustrate that it is preferable to tackle simplification in entry and bookkeeping costs than large tax cuts, as they can increase firm entry and represent smaller losses in terms of revenues. However, policymakers must be careful about what they expect as the benefits of such reductions. In the Brazilian setting, benefits mainly came from an increase in social security coverage, with no large observable effects on the activity of these microentrepreneurs.

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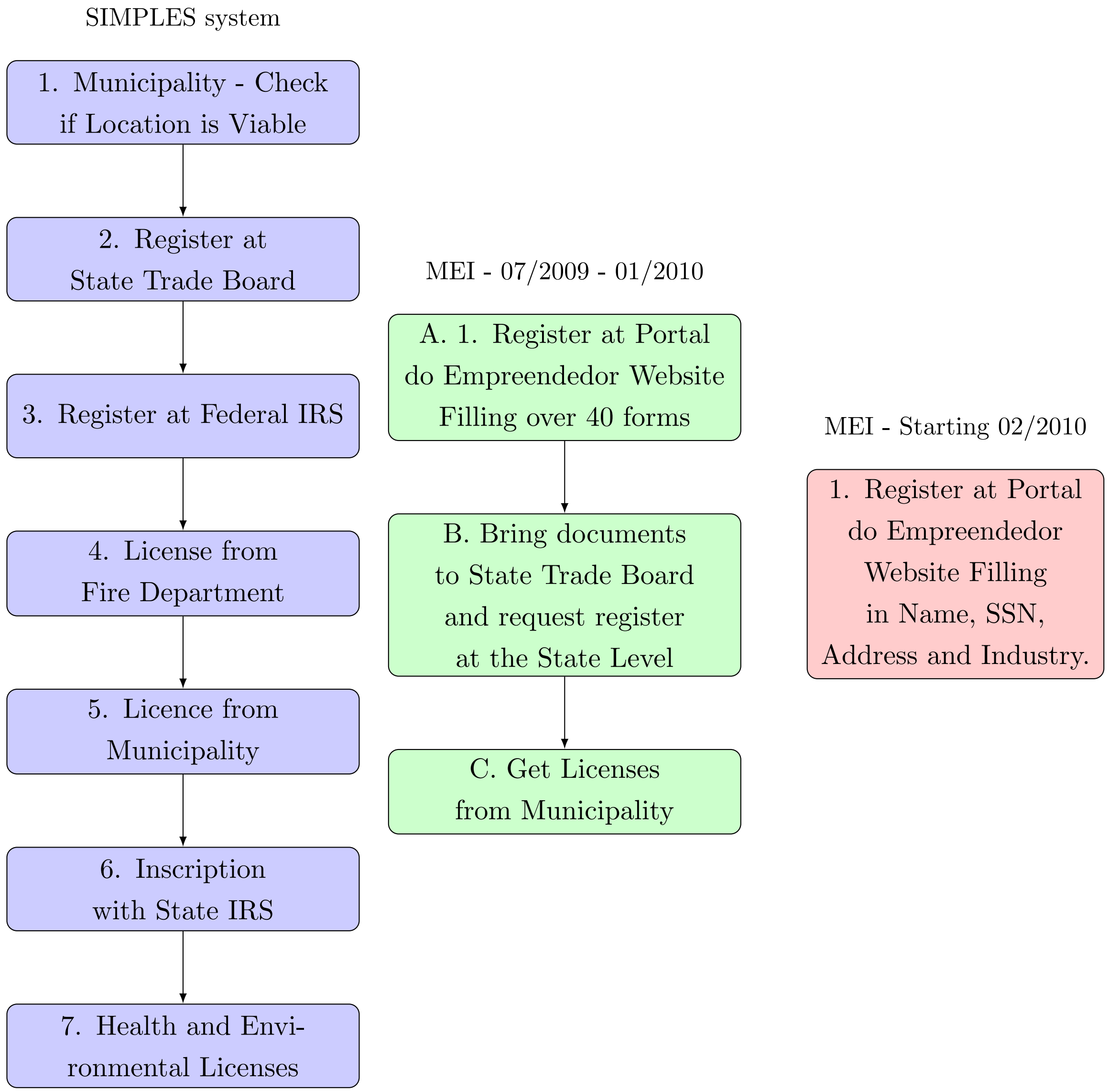
Table of Contents - Appendix

- Additional Figures
- Additional Tables
- **Appendix A:** Additional Data Description
- **Appendix B:** Compliers' Characteristics Adaptation of Abadie (2003)

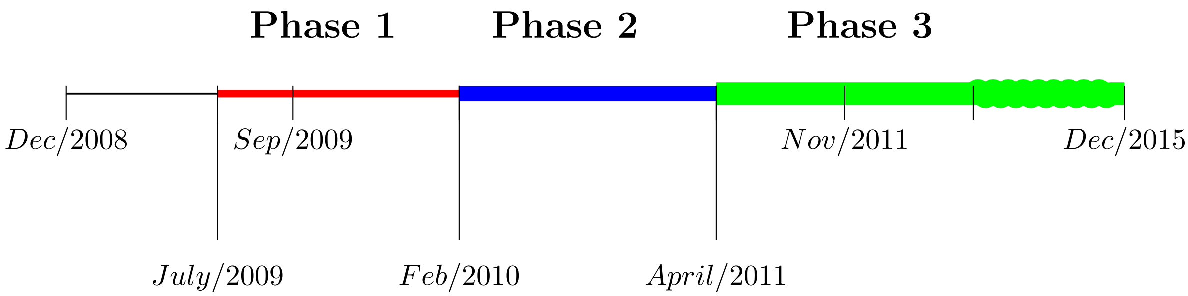
Additional Figures

Figure A1: Steps Necessary to Register a Firm



This figure summarizes the process to register a firm in the SIMPLES system, and in the MEI system, before and after the creation of the integrated system in February 2010.

Figure A2: Timeline of The MEI program



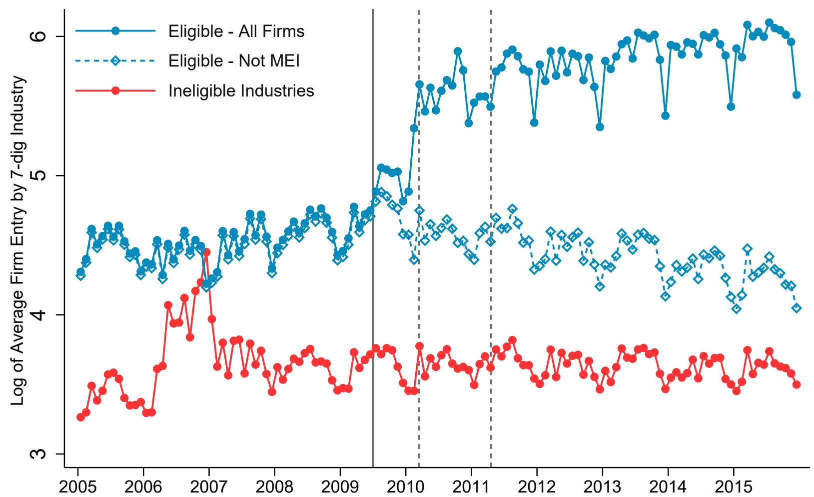
Event Details:

- **December 2008:** The project was sanctioned by the federal government.
- **July 2009: Beginning of Phase 1 (Implementation Phase)** Implementation began in a selected group of states (SP, RJ, DF, MG), and the integrated website is not ready yet.
- **September 2009:** another group of states were allowed (CE, ES, PR, RS, SC), integrated website was not ready yet.
- **February 2010 Beginning of Phase 2:** All states included and integrated website started to work.
- **April 2011 Beginning of Phase 3:** Tax rate reduced from 11% of the Minimum Wage to 5%.
- **November 2011:** Eligibility revenue cap raised to 60,000 R$.

Short Summary of Formalization Costs at each Phase:

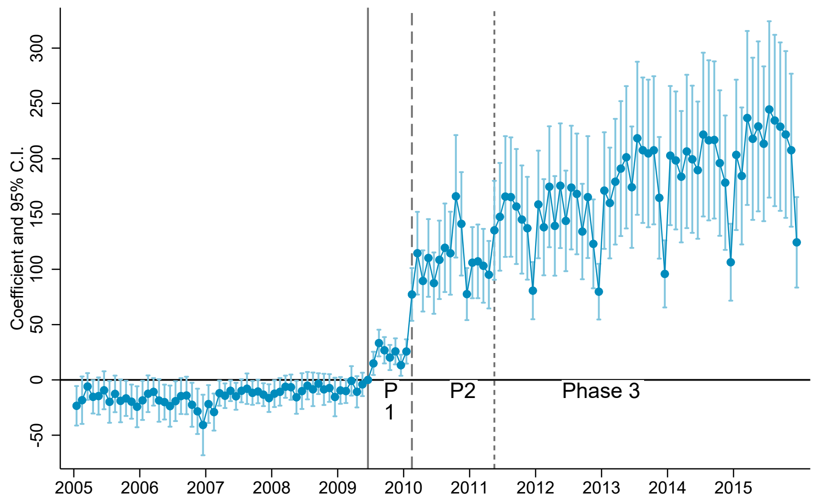
- **Phase 1:** Tax rate at 11% of Minimum Wage. Registration costs are lowered, but the integrated website is not working and is still costly to register firms.
- **Phase 2:** Tax rate still at 11% of Minimum Wage. Registration costs were virtually eliminated, and the program worked throughout the whole country.
- **Phase 3:** Tax rate at 5% of Minimum Wage. Registration costs were virtually eliminated, and the program worked throughout the whole country.

Figure A3: Average Log of Firm Registration By Industry Eligibility



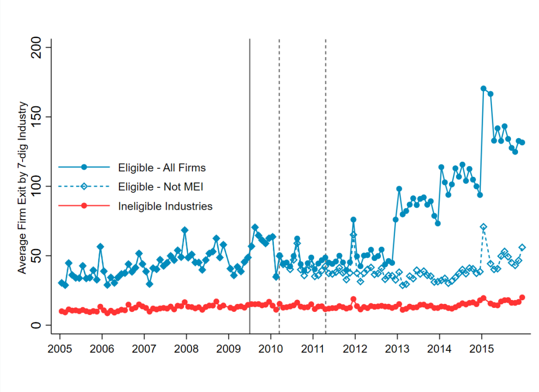
Notes: This figure plots the log of average firm registration by industry for eligible and ineligible 7 digit CNAE classification. List of eligible industries were defined from the *CGSN 58* at the 7 digit CNAE. The blue connected line with solid markers shows the average firm registration in eligible industries for firms in all types of tax systems (MEI and regular firms). The blue connected line with hollow markers shows the average firm registration of firms not in the MEI program for industries in the eligible sectors. The red line shows average firm registration in ineligible industries. Vertical lines represent the start of the different phases of the program.

Figure A4: Difference in Differences Coefficients on Firm Entry - Levels

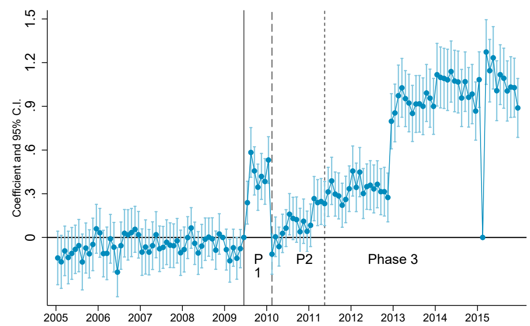


This figure plots the vector of coefficients β_t and 95% Confidence Intervals from equation 1. The sample consists of a monthly panel at the 7-digit industry level from 2005 to 2015. The outcome is the number of firms entering each period. Vertical lines represent the start of the different phases of the program. Standard errors are clustered at the 7-digit CNAE level.

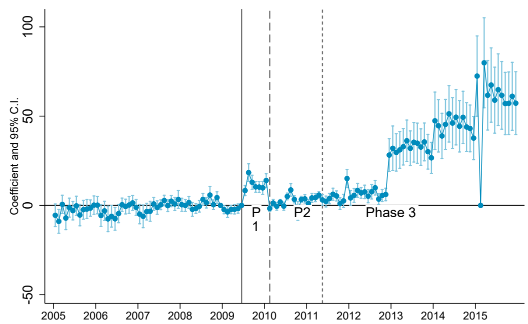
Figure A5: Effects on Firm Exit



(a) Average by Industry



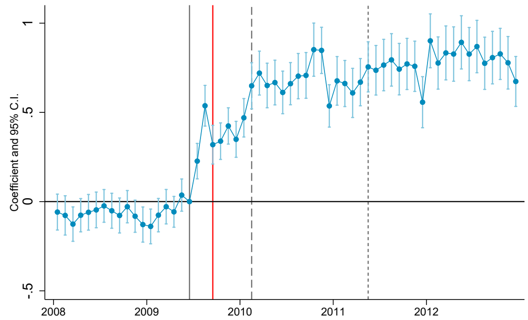
(b) Estimates on Log of Firm Exit



(c) Estimates on Levels of Firm Exit

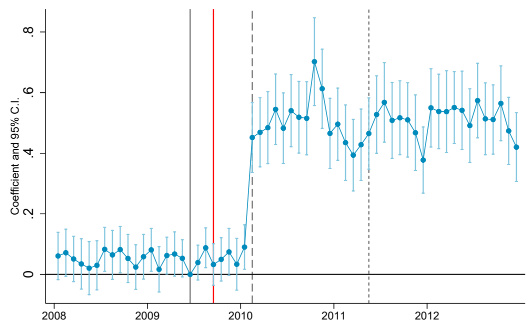
Notes: Panel A plots the average firm exit by industry for eligible and ineligible 7 digit CNAE classification. List of eligible industries were defined from the *CGSN 58* at the 7 digit CNAE. The blue connected line with solid markers shows the average firm exit in eligible industries for firms in all types of tax systems (MEI and regular firms). The blue connected line with hollow markers shows the average firm exit of firms not in the MEI program for industries in the eligible sectors. The red line shows average firm exit in ineligible industries. Vertical lines represent the start of the different phases of the program. Panels B and C plots the estimates of equation 1 on both logs and levels of firm exit.

Figure A6: Difference in Differences Coefficients on Log of Firm Entry - by State



(a) SP, RJ, MG, DF

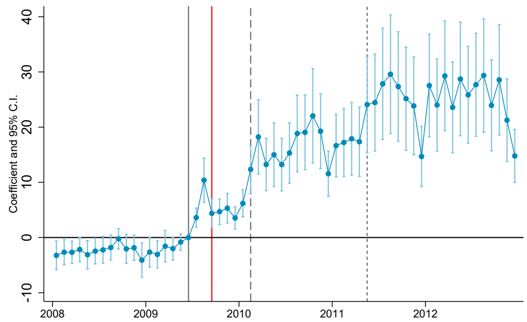
(b) RS, SC, PR, CE



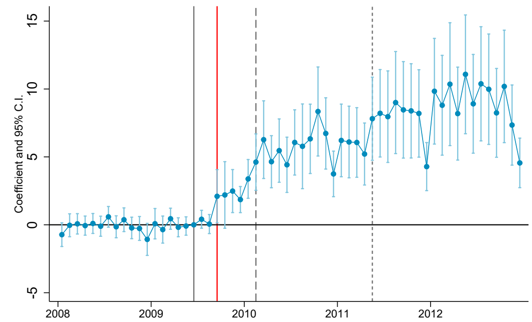
(c) Remainder States

This figure plots the vector of coefficients β_t and 95% Confidence Intervals from equation 1. The sample consists of a monthly panel at the 7-digit industry level from 2008 to 2012. The outcome is the log of the number of firms entering each period. Vertical gray lines represent the start of the different phases of the program. Vertical red line represents September 2009 Standard errors are clustered at the 7-digit CNAE level. SP, RJ, MG and DF were allowed into the program in July 2009, RS, SC, PR and CE were allowed in September 2009, and remainder states were allowed in February 2010.

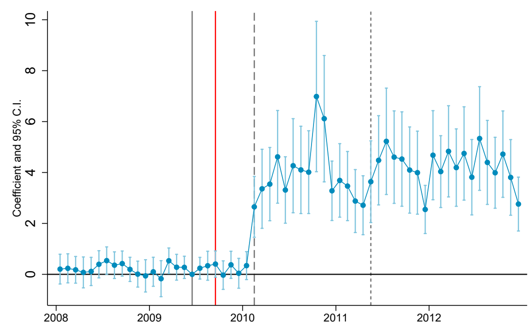
Figure A7: Difference in Differences Coefficients on Firm Entry - Levels by State



(a) SP, RJ, MG, DF



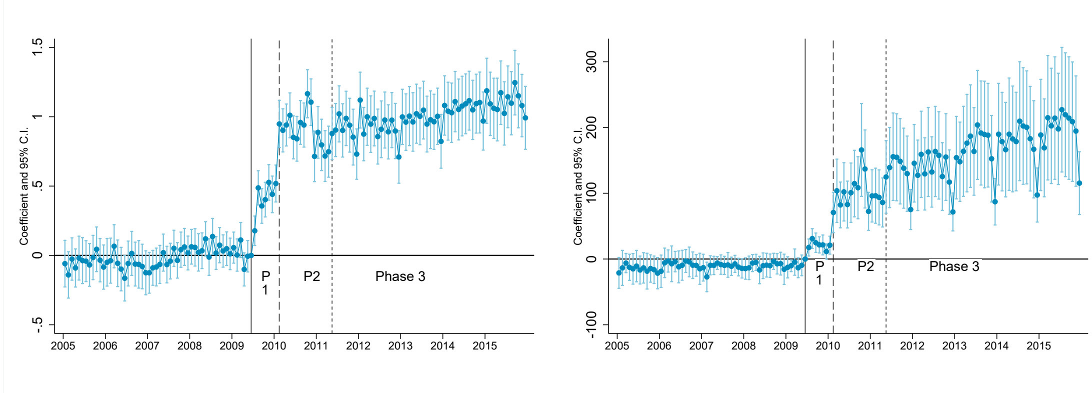
(b) RS, SC, PR, CE



(c) Remainder States

This figure plots the vector of coefficients β_t and 95% Confidence Intervals from equation 1. The sample consists of a monthly panel at the 7-digit industry level from 2008 to 2012. The outcome is the number of firms entering each period. Vertical gray lines represent the start of the different phases of the program. Vertical red line represents September 2009 Standard errors are clustered at the 7-digit CNAE level. SP, RJ, MG and DF were allowed into the program in July 2009, RS, SC, PR and CE were allowed in September 2009, and remainder states were allowed in February 2010.

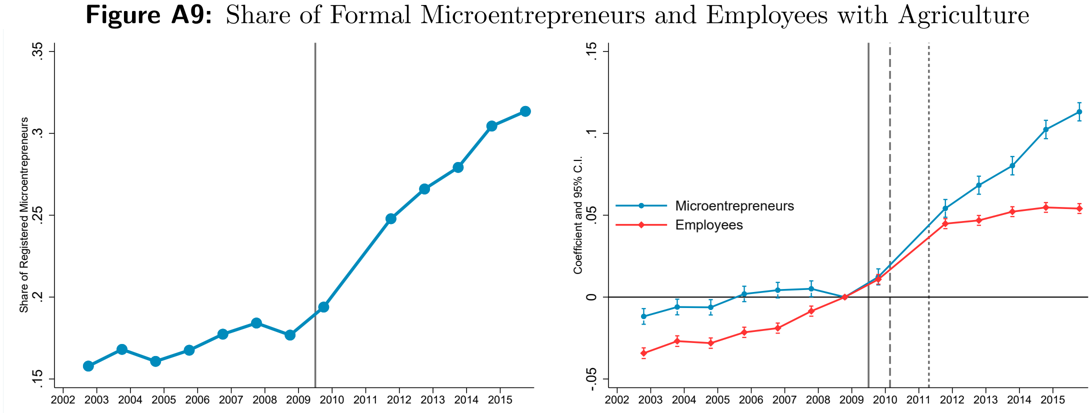
Figure A8: Difference in Differences Coefficients on Firm Entry - Matched Sample



(a) Log of Firm Entry **(b)** Levels of Firm Entry

This figure plots the vector of coefficients β_t and 95% Confidence Intervals from equation 1. The sample consists of a monthly panel at the 7-digit industry level from 2005 to 2015 using industries matched. Match is done without replacement using a propensity score matching that estimates propensity to treatment using 2008 firm entry, number of active firms, and number of firms in the SIMPLES system. The outcome is the log of (Panel A) and the levels (Panel B) of the number of firms entering each period. Vertical lines represent the start of the different phases of the program. Standard errors are clustered at the 7-digit CNAE level.

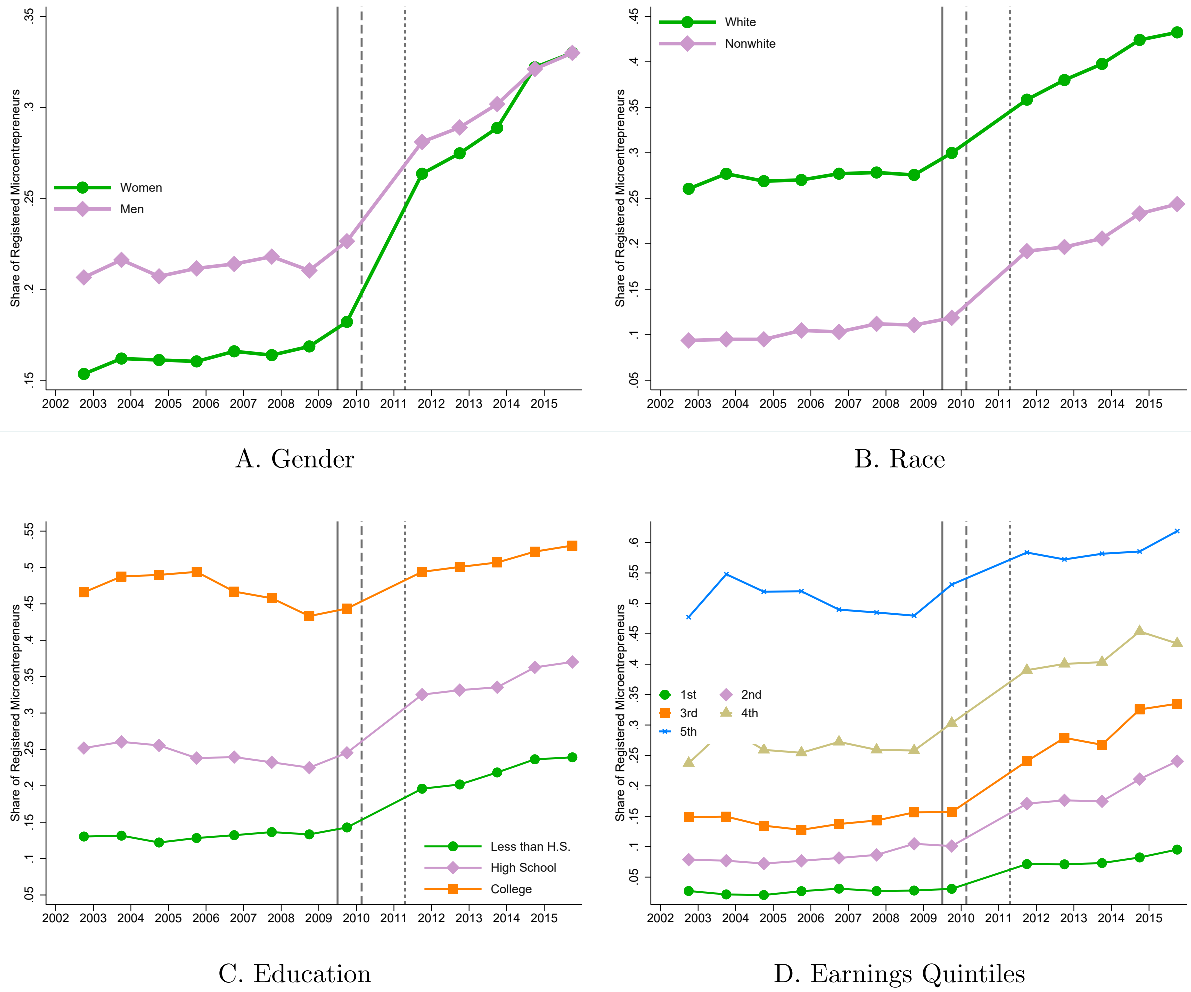
Figure A9: Share of Formal Microentrepreneurs and Employees with Agriculture



(a) Microentrepreneurs **(b)** Controlling by Observables

Microentrepreneurs are defined as individuals who declare themselves as self-employed or employers who hire at most one employee. We define a formal microentrepreneur as those who declare to be contributing to social security through their job. Panel A plots the share of microentrepeneurs between 18 and 60 years that are formal by year. Panel B plots the coefficients α_t of two separate regressions. The blue one, is with the sample of microentrepreneurs. The red one with the sample of individuals working as employees of a firm. All results were built with PNAD 2002 - 2015.

Figure A10: Share of Formal Microentrepreneurs by Group



A. Gender

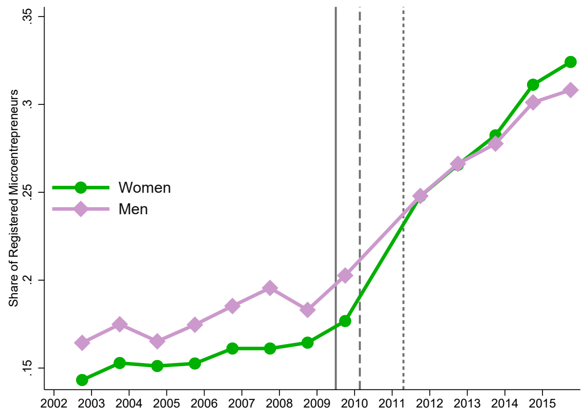
B. Race

C. Education

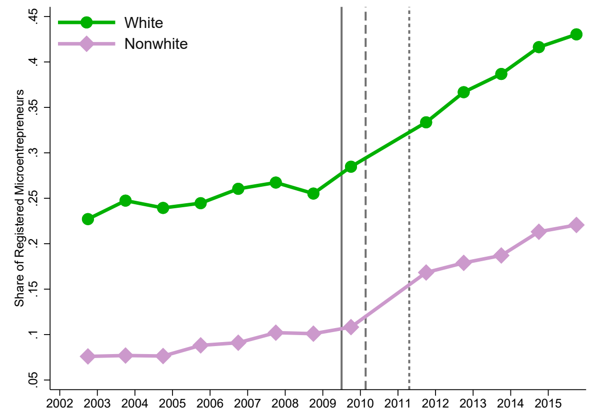
D. Earnings Quintiles

Notes: All panels are built from PNAD data set. Sample is restricted to microentrepreneurs, which are defined as individuals that declare themselves self-employed or firm owners with at most one employee. Sample is restricted to individuals between 18 and 60 years. Panel A divides microentrepreneurs in three broad education groups by the highest degree that they completed: Less than High School, High School and More than High School. Panel B divides the sample in earnings quintiles. Panel C plots the share of microentrepreneurs that are formal by gender. Panel D plots the share of microentrepreneurs years that are formal by race. Vertical lines represent the start of the different phases of the program.

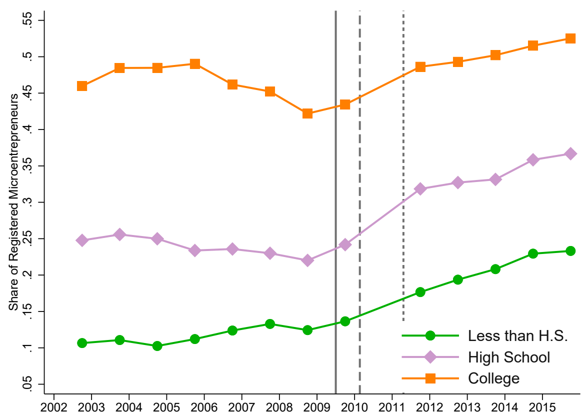
Figure A11: Share of Formal Microentrepreneurs by Group including Agriculture



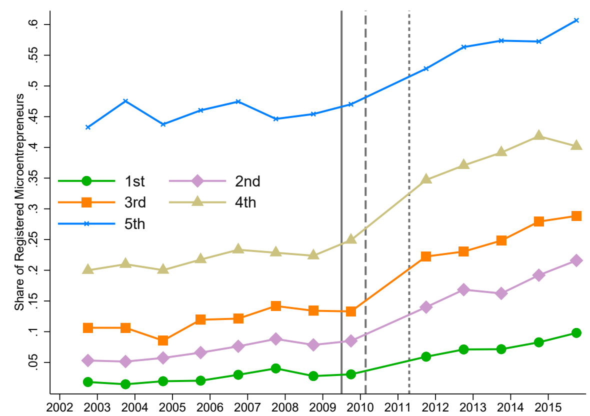
A. Gender



B. Race



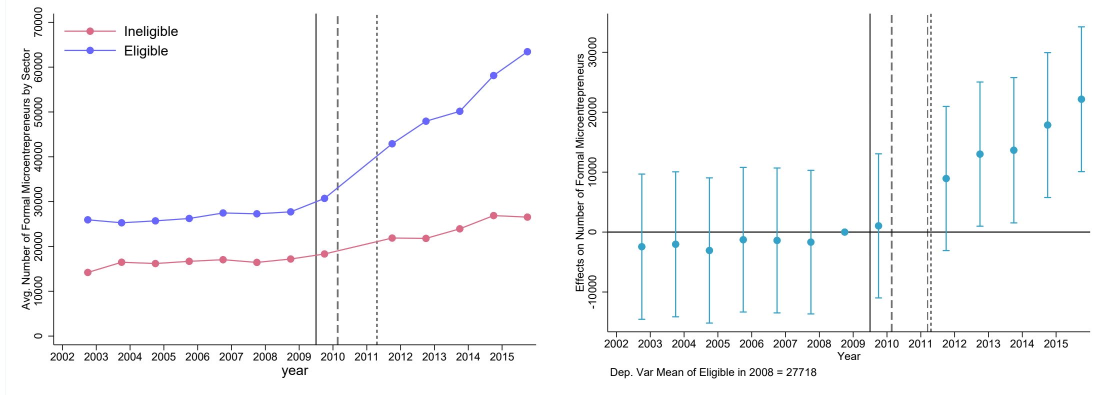
C. Education



D. Earnings Quintiles

Notes: All panels are built from PNAD data set. Sample is restricted to microentrepreneurs, which are defined as individuals that declare themselves self-employed or firm owners with at most one employee. Sample is restricted to individuals between 18 and 60 years. Panel A divides microentrepreneurs in three broad education groups by the highest degree that they completed: Less than High School, High School and More than High School. Panel B divides the sample in earnings quintiles. Panel C plots the share of microentrepreneurs that are formal by gender. Panel D plots the share of microentrepreneurs years that are formal by race. Vertical lines represent the start of the different phases of the program.

Figure A12: Effects on Formalization of Microentrepreneurs

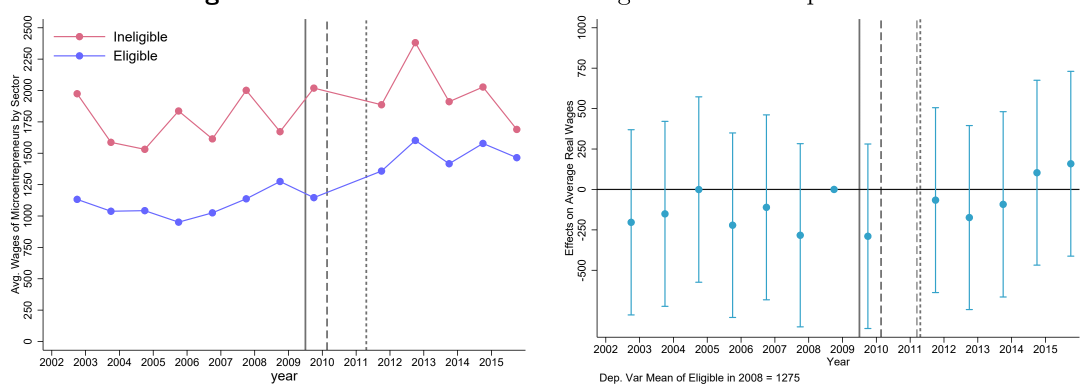


(a) Formal Microentrepreneurs - Trends

(b) Formal Microentrepreneurs

This figure shows formalization of microentrepreneurs across eligible and ineligible industries using PNAD data. Panel (a) shows the number of formal microentrepreneurs among eligible and ineligible industries. Panel (b) plots the coefficients β_t from equation 5. The sample comprises all microentrepreneurs in PNAD between 18 and 60 years old. We include as controls the share of women, the share of nonwhite, average years of schooling, average hours worked, and share of microentrepreneurs in each state.

Figure A13: Effects on Labor Earnings of Microentrepreneurs

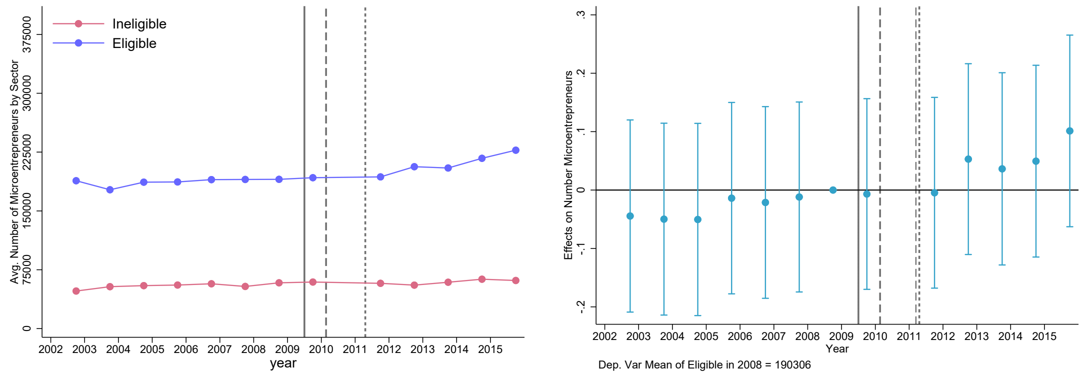


(a) Average Labor Earnings - Trends

(b) Average Labor Earnings

This figure shows the average labor earnings of microentrepreneurs across eligible and ineligible industries using PNAD data. Labor Earnings are adjusted to December 2018 values. Panel (a) shows the number of formal microentrepreneurs among eligible and ineligible industries. Panel (b) plots the coefficients β_t from equation 5. The sample comprises all microentrepreneurs in PNAD between 18 and 60 years old. We include as controls the share of women, the share of nonwhite, average years of schooling, average hours worked, and share of microentrepreneurs in each state.

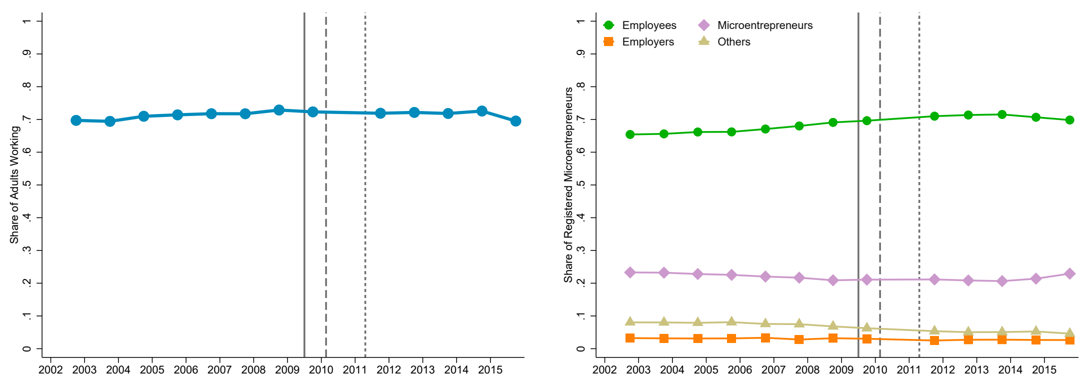
Figure A14: Effects on Number of Microentrepreneurs



(a) Number of Microentrepreneurs - Trends **(b)** Number of Microentrepreneurs

This figure shows the average number of microentrepreneurs across eligible and ineligible industries using PNAD data. Panel (a) shows the number of formal microentrepreneurs among eligible and ineligible industries. Panel (b) plots the coefficients β_t from equation 5 divided by the average of the outcome variable in 2008. The sample comprises all microentrepreneurs in PNAD between 18 and 60 years old. We include as controls the share of women, the share of nonwhite, average years of schooling, average hours worked, and share of microentrepreneurs in each state.

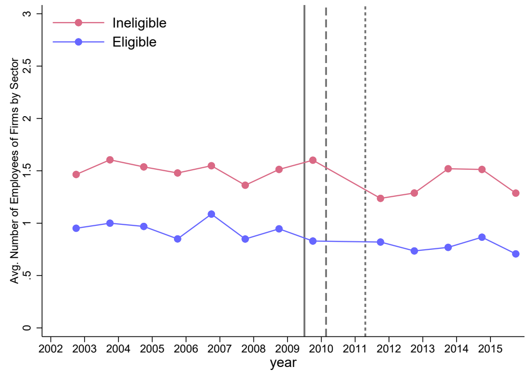
Figure A16: (Lack of) Occupational Change



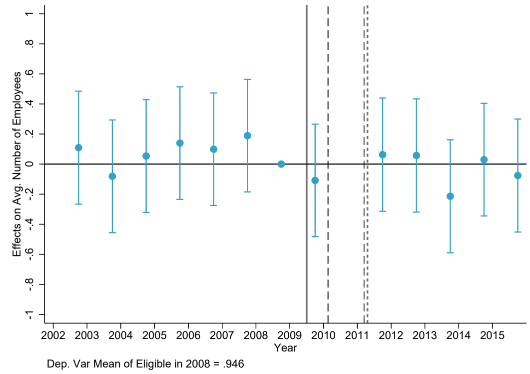
(a) Share Working **(b)** Share by Occupation Category

Panel A plots the share of individuals between 18 and 60 years that are working. Panel B plots the share of workers between 18 and 60 years in each type of occupation. Microentrepreneurs are defined as individuals that declare themselves self-employed or firm owners with at most one employee. Employers are firm owners that have more than one employee. Employees include formal and informal workers that work for a monetary income in a firm or in the public sector. Others category include workers that work for non monetary benefit which mostly encompasses farmers in that work for their own consumption. Vertical lines represent the start of the different phases of the program.

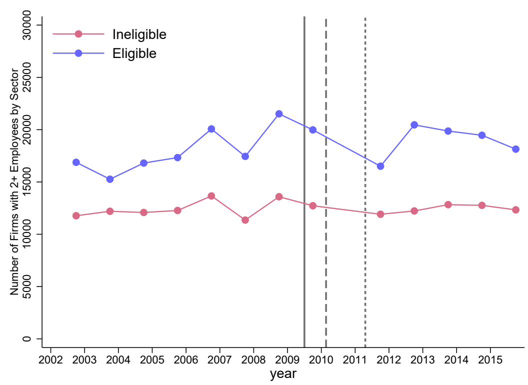
Figure A15: Effects on Firm Size



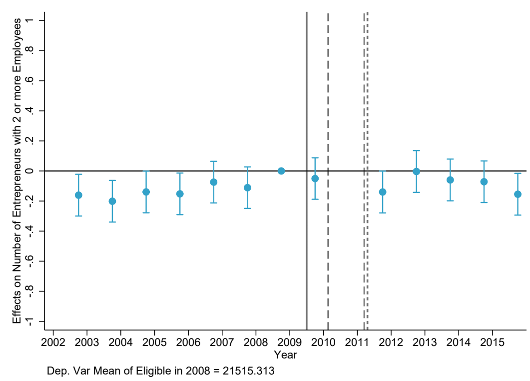
(a) Avg. Number of Employees - Trends



(b) Avg. Number of Employees



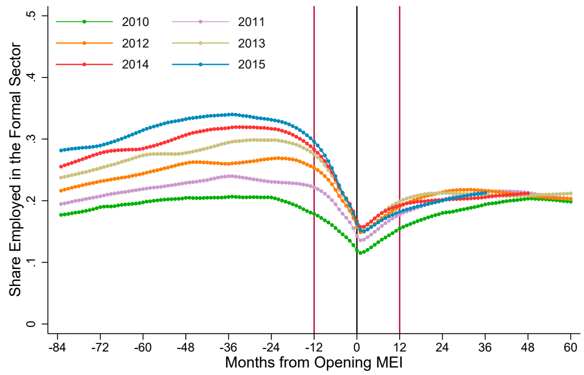
(c) Entrepreneurs with ≥ 2 Workers



(d) Number of Entrepreneurs with ≥ 2 Workers

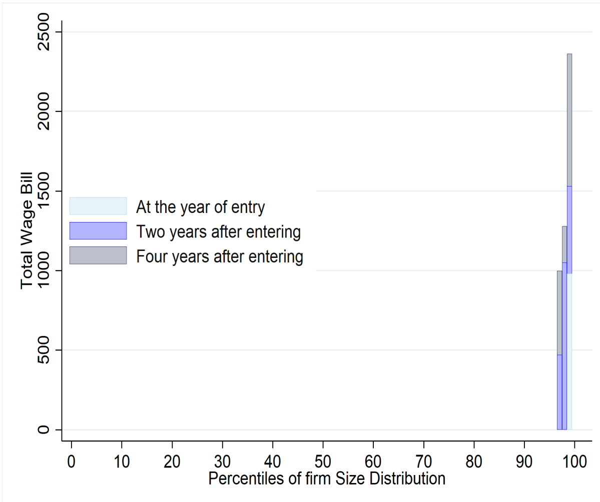
This figure shows results on firm size across eligible and ineligible industries using PNAD data. Panel (a) shows the average number of employees of firms in eligible and ineligible industries. Panel (c) shows the number of employers with 2 or more employees in PNAD. Panels (b) and (d) plots the coefficients β_t from equation 5 divided by the average of the outcome variable in 2008. The sample comprises all self employed or employers in PNAD between 18 and 60 years old. We include as controls the share of women, the share of nonwhite, average years of schooling, average hours worked, and share of microentrepreneurs in each state. The number of employees is a categorical variable in RAIS. We use the minimum of each category as the input for the number of employees

Figure A17: Share of Microentrepreneurs that were employed in the formal Sector

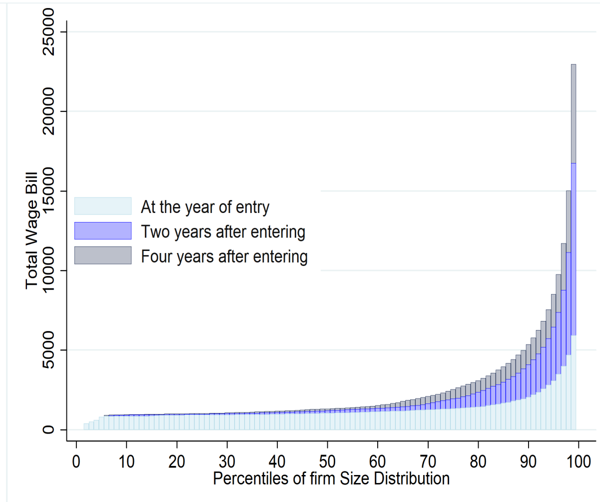


Notes: This figure plots the share of individuals that were employed in the formal sector relative to the month at which they opened their MEI by the year they opened their firm. The sample comprises all microentrepreneurs with a valid Social Security Number in the Tax Authority administrative records that registered between Feb. 2010 and Dec. 2015. Being employed in the formal sector means having a valid contract registered at RAIS. Vertical red lines mark one year before and after the registration as a MEI and the black line marks the month at which the individual registered as a MEI.

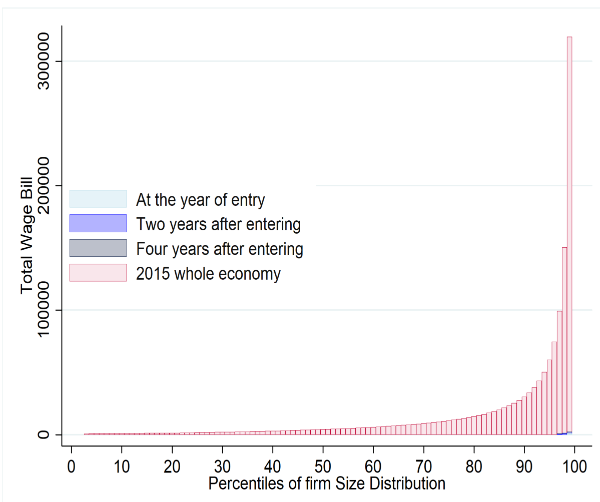
Figure A18: Wage Bill Distribution of Microentrepreneurs that Register as MEI



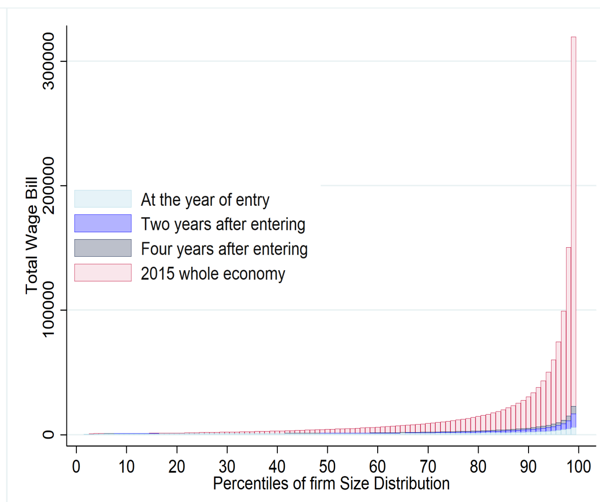
A. Wage Bill Unconditional



B. Wage Bill - Conditional



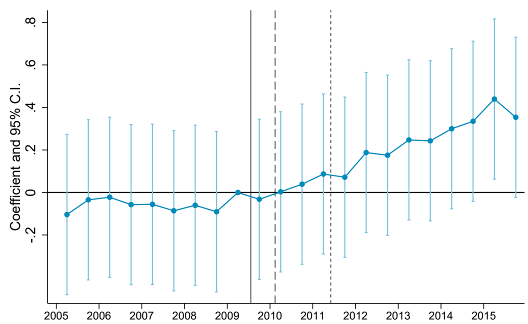
C. Unconditional with RAIS



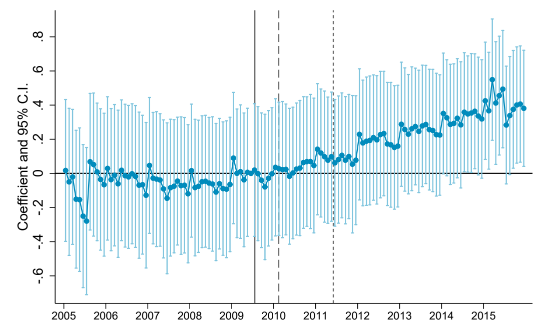
D. Conditional With RAIS

Notes: This figure is built from the panel of MEI firms combined with ownership data and the economy's distribution of firm size from RAIS. Firm size is calculated as the total number of individuals that were reported in RAIS as formally employed by December 31st at the respective firm. On panel (A), we input zero workers to an active MEI firm that does not appear in RAIS. On panel (B) we restrict our sample to microentrepreneurs that do appear in RAIS. Panels (C) and (D) are the same as in (A) and (B) but adding the observed economy Firm Size distribution in RAIS in 2015.

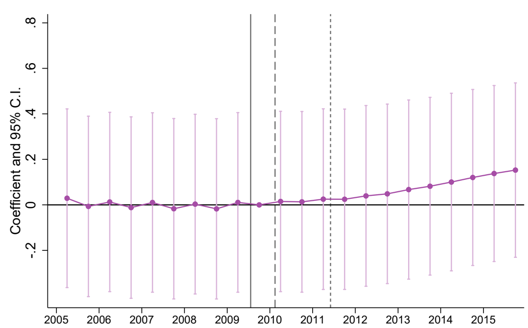
Figure A19: Effects on the Log of the Number of Benefits Issued



(a) Semester Level



(b) Month Level

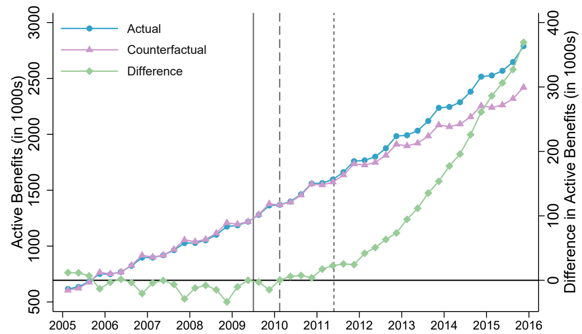


(c) Semester Level

(d) Month Level

This Figures show estimates β_t from equation 6. Panels (a) and (b) use as the outcome the log of benefits issued, whereas Panels (c) and (d) shows the log of benefits that are active in the given quarter. The benefit-level data is aggregated at the State x Month x Class level in Panels (a) and (c) and at the State x Month x Class level in Panels (b) and (d). Vertical lines indicate the start of the different phases of the program.

Figure A20: Estimates of Total Benefits Active to Individual Contributors



Notes: This figure plots our estimates of the actual and counterfactual benefits for Individual Contributors in the Social Security data. In the right y-axis we plot the difference between both measures.

Additional Tables

Table A1: Summary Statistics of the MEIs in the tax authority Data

	< 2010	2010	2011	2012	2013	2014	2015	Total
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Total entry of MEI's								
	214,805	743,041	922,981	1,058,196	1,255,788	1,342,726	1,488,381	7,025,918
Panel B: Share of MEI's that have a missing Social Security Number								
	191,711	50,646	56,064	55,390	52,660	46,389	44,360	497,220
	0.892	0.068	0.061	0.052	0.042	0.035	0.030	0.071
Panel C: Share of MEI's that were active for at least 1 year								
	213,275	723,856	897,873	993,290	1,120,888	1,181,320	1,285,871	6,416,373
	0.993	0.974	0.973	0.939	0.893	0.880	0.864	0.913
Panel D: Share of MEI's by Region								
North	1,912	58,537	54,132	61,807	68,743	68,689	73,632	387,452
	0.009	0.079	0.059	0.058	0.055	0.051	0.049	0.055
Northeast	38,587	184,756	189,823	204,392	241,135	250,868	272,239	1,381,800
	0.180	0.249	0.206	0.193	0.192	0.187	0.183	0.197
Southeast	111,066	326,451	467,924	535,902	636,425	694,346	780,020	3,552,134
	0.517	0.439	0.507	0.506	0.507	0.517	0.524	0.506
South	49,981	98,657	127,764	160,027	194,951	209,978	236,686	1,078,045
	0.233	0.133	0.138	0.151	0.155	0.156	0.159	0.153
Midwest	13,259	74,639	83,338	96,068	114,534	118,845	125,804	626,487
	0.062	0.100	0.090	0.091	0.091	0.089	0.085	0.089

Notes: Table elaborated by the authors from the tax authority data set. A unique firm is defined by an unique CNPJ. Regions are divided according to the official geographic division of the country. A firm is defined as active if its official record hasn't changed to inactive or suspended.

Table A2: Estimates of Program's Effect on Firm Entry

	(1)	(2)	(3)	(4)
	Log(Firm Entry)		Firms Entry	
Eligible x Phase 1	.43***	.43***	52***	32***
	(.034)	(.042)	(9.3)	(8.7)
Eligible x Phase 2	.96***	.92***	172***	112***
	(.052)	(.073)	(30)	(26)
Eligible x Phase 3	1.1***	1***	249***	176***
	(.067)	(.097)	(45)	(39)
Dep. Var. Mean of Treated	3.04	3.06	94.02	94.88
Month x 2 digit Ind. F.E.	No	Yes	No	Yes
Observations	80800	79278	80800	79278

This table presents estimates of coefficients $\{\beta_1, \beta_2, \beta_3,\}$ from equation 2 using a sample of matched CNAE industries. The sample consists of a monthly panel at the 7-digit industry level from 2005 to 2015 using industries matched. Match is done without replacement using a propensity score matching that estimates propensity to treatment using 2008 firm entry, number of active firms, and number of firms in the SIMPLES system. Columns (1) and (2) use as outcome the log of firm registration at a given month for a given 7-digit industry. Columns (3) and (4) show results using the number of firms registering as the outcome. Standard Errors are clustered at the industry level.

Table A3: Summary Statistics of MEI work Activity

	2011	2012	2013	2015
	(1)	(2)	(3)	(4)
Location of Work:				
Works at: Street	0.180	0.120	0.104	0.130
Works at: Office	0.390	0.340	0.302	0.280
Works at: Home	0.400	0.430	0.486	0.530
Works at: other's Firms	.	0.110	0.107	0.0700
Works at: Other (2011)	0.0300	.	.	.
Sector of Work:				
Sector: Construction	0.0730	0.0800	0.0880	0.0950
Sector: Retail	0.395	0.390	0.393	0.374
Sector: Manufacturing	0.176	0.170	0.147	0.153
Sector: Services	0.356	0.360	0.367	0.372
Motive for Registering as MEI				
Motive: Formal Firm Benefits	0.630	0.690	0.785	0.680
Motive: Social Security Benefits	0.370	0.310	0.215	0.320
Had Other Income Sources	0.220	0.260	0.240	0.230
Where you active as MEI at the moment of the survey				
Yes	-	0.90	0.83	0.88
No	-	0.10	0.17	-
No - Ended Business	-	-	-	0.08
No - Still haven't Started	-	-	-	0.03
No - Outgrown MEI	-	-	-	0.01
Observations	10585	11577	12000	9657

Notes: This table is elaborated from information extracted from Perfil do MEI surveys. Their sample is a random sample from individual microentrepreneurs registered at the federal tax authority.

Table A4: Firm Size distribution of Microentreprises.

	(1)	(2)	(3)	(4)	(5)	(6)
Distribution of firm size t years after opening						
	At t = 0	t = 1	t = 2	t = 3	t = 4	t = 5
Mean	0.0197	0.0409	0.0553	0.0666	0.0811	0.0918
10th Percentile	0	0	0	0	0	0
Median	0	0	0	0	0	0
75th Percentile	0	0	0	0	0	0
90th Percentile	0	0	0	0	0	0
95th Percentile	0	0	0	0	0	0
99th Percentile	1	1	1	1	2	2
Observations	6516136	6052015	5694839	5376262	3861606	2769368

Distribution of firm size t years after opening						
	At t = 0	t = 1	t = 2	t = 3	t = 4	t = 5
Mean	1.177	1.511	1.776	2.006	2.150	2.272
10th Percentile	1	1	1	1	1	1
Median	1	1	1	1	1	1
75th Percentile	1	1	1	2	2	2
90th Percentile	2	2	3	4	4	4
95th Percentile	2	4	5	5	6	6
99th Percentile	4	8	11	13	14	15
Observations	109018	164010	177379	178416	145618	111876

Notes: This table was made from the panel of MEI firms combined with ownership data. Firm size is calculated as the total number of individuals that were reported in RAIS as formally employed by December 31st at the respective firm.

Table A5: Effects on Social Security Benefits - Type of Benefit

	(1) All	(2) Health	(3) Maternity	(4) Pensions	(5) Retirement
Eligible x Phase 1	.024	-.086	.35*	-.0032	.038
	(.11)	(.084)	(.2)	(.096)	(.12)
Eligible x Phase 2	.096	.024	.51***	-.008	.02
	(.072)	(.051)	(.14)	(.067)	(.077)
Eligible x Phase 3	.31***	.11***	.96***	.04	.12**
	(.044)	(.034)	(.085)	(.04)	(.048)
Dep. Var. Mean of Treated	7.35	6.63	4.88	5.17	6.11
Observations	2376	2376	2370	2376	2376

Notes: This table presents estimates of coefficients $\{\beta_1, \beta_2, \beta_3\}$ from equations 7. Column (1) replicates the main estimates using all benefits. Columns (2) use as outcome benefits given due to health reasons. Column (3) shows results using as outcome maternity benefits. Column (4) show results using pensions and Column (5) retirement benefits. The benefit-level data is aggregated at the State x Month x Class.

A Additional Data Description

In this appendix, we describe the details of the CNPJ data used in our analysis and thoroughly discuss the topic as the major difference between this project and previous research on the subject.

The CNPJ data

The *Cadastro Nacional da Pessoa Jurídica* (CNPJ) is the administrative database under the responsibility of *Receita Federal do Brasil* (RFB) that holds registration information of entities (e.g., business partnerships, individual entrepreneurs, public agencies, among others) of interest to the tax administrations of the federal and local governments in Brazil. It is updated constantly by the *Coordenação Geral de Gestão de Cadastros* (Cocad), the coordination responsible for taking the administrative procedures to publicly notify entities with register irregularities. And it is maintained by the *Serviço Federal de Processamento de Dados* (SERPRO), a data processing public company that provides technology services to the Brazilian federal government.

The CNPJ database has two parts: registration information of the entities in CNPJ; and, registration information of partners and administrators of some types of entities in the database. It used to be a confidential dataset to which access had to be granted by RFB which hardly ever authorized it. Eventually, the matter became judicial and the *Controladoria Geral da União* (CGU) ended up deciding - in late 2018 - in favor of the claimants, determining that the database should be open to all Brazillians[42]. Ever since then, the RFB publishes quarterly, in its website, the last version of the data processed by SERPRO withholding the CPF numbers of the partners and administrators in the dataset.

Sample Restrictions

In our analysis, we leave out the partners and administrators' information and make use only of the entities' registration data. The entities registered in CNPJ are not restricted to business units, in fact, they are of significantly different types. According to *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 3 and 4, the entities obliged to register in CNPJ include Business units, Entrepreneurs, States and Cities, Public agencies, Condominiums buildings, Groups and consortia of companies, Employers' consortia, Clubs and investment funds, Foreign diplomatic representations in Brazil, Diplomatic representations of the Brazilian State abroad, Permanent representations of international organizations or extraterritorial institutions in Brazil, Notary and registration services, Public and Private Funds, Political candidates running for elective office , Plebiscitary and referendum fronts, Real state developments under special taxation regime, Polinational commissions created by an international act between Brazil and other countries, International entities that hold real or financial assets in

[42]See http://dados.gov.br/noticia/cgu-libera-acesso-a-base-de-dados-de-cnpj-da-rfb .

Brazil, Foreign bank institutions trading Brazilian Real currency, Partnerships in Membership Account and others.

Thus, in order to perform an accurate analysis on entrepreneurship and firm creation, we sort out the several non-business related entities in the data. We make use of Concla/IBGE's legal nature classification, one of the native variables available in CNPJ, to keep only a subset of the businesses entities that corresponds to individual entrepreneurs and private, for profit, business partnerships endowed with legal personality under private law (Table A6). After all the restrictions, we still have around 70% of the original establishments. Besides, our main results such as figure 2 hold if we use the full sample, with the exception of the month prior to the elections where there is an enormously large number of political campaign headquarters registering.

It has information as detailed as entity's full address, telephone number, email and more. Table A7 lists the native variables used in our analysis. It includes an indicator variable of participation in the MEI program, the entity's date of start of activity, a 7-digit industry code, the Brazilian state of location, the already mentioned legal nature code, the legal name of the entity, the entity's current registration status and the date of registration status change.

Data Quality - Firm Activity

Asides regulatory and fiscal obligations, entities are obliged to maintain registration data up to date in Brazil[43]. Cocad has the authority to restrain entities' registration statuses from being active in case of repeated failure to comply with fiscal obligations, inconsistencies in the registration information, amongst several other occasions[44]. For instance, if an entity fails to present any mandatory statement for two consecutive years[45] or fails to confirm receiving two or more tracked mail from RFB, the Cocad can publicly summon the entity by notice and set its registry status as inapt. An important example rather not related to compliance failure is when an entity is in the process to close its register with RFB, its registry status is set to suspended throughout the whole operation.

On such occasions, Cocad not only updates the entities' registry statuses but also records the date of the change in CNPJ as well. This fact, along with the entities' date of start of the activity, provided us the beginning and the end of activity spell of each entity in CNPJ which allowed us to filter them in regard to such information. This way, we only keep in our sample the entities that were active between January 2005 and December 2015 in CNPJ.

A register can be closed either by the entity's responsible requests or administratively, by the RFB. There are five situations for a company to get administratively closed by the RFB: notified by repeated omission; notified by administratively inexistance; inapt not regularized in 5 years; with cancelled registration by the register agency; or, by court order. The former three

[43] *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 24.

[44] See *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 38-51.

[45] *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 29, item I.

occur when detected lack of comply of obligations. Default omission is the entity that, being obliged to, has not submitted, for 5 (five) or more years, none of the statements.

The CNPJ register is suspended[46] mainly when: the entity's responsible place a closing request to RFB and it is either under analysis or if it has been rejected; it is declared legally inexistent[47]; temporarily interrupted its activities and have declared this situation to RFB; it is declared legally an insistently non-complier, which is one that, being obliged to, has not submitted, for 5 (five) or more years, none of the statements[48]; or, have inconsistencies in the registration data[49]. In the last two cases, the Cocad publicly notifies the entities, that has to regularize their situation with RFB within a specific time, otherwise the register is closed. Thus there is an active administrative control over the registries in the database, giving some reliability to the data.

The CNPJ register gets inapt when: omission of declarations and statements, thus considered the one that, being obliged, fails to present, in 2 (two) consecutive years, any of the declarations and statements[50] ; failed to receive 2 tracked correspondences from RFB; with irregularities in foreign trade operations, thus considered one that does not prove the origin, availability and effective transfer, if applicable, of the resources used in foreign trade operations, as provided by law. In the case of an inapt register, the entity suffers sanctions such as it is included in the Informative Register of Unpaid Credits of the Federal Public Sector (Cadin); it is prevented from participate in public purchase auctions, enter into covenants, agreements, adjustments or contracts that involve disbursement, in any capacity, of public resources, and respective amendments; obtain tax and financial incentives; carry out credit operations that involve the use of public resources; and, transact with banking establishments, including with regard to the movement of current accounts, the realization of financial investments and the obtaining of loans. Hence, there are incentives for the responsible to keep the registration information up to date so they do not get their registered inapt.

Other Registration statuses are active, suspended, inapt and close. We use the date of opening and the date of change in registration status to other than active to determine the establishments that were active in a determined time frame. We keep only the establishments active between 2005 and 2015.

Data Accuracy

The law provision determining the responsibility on the registration data accuracy to the entity is not the only existent instrument to enforce compliance towards registration obligations. Outdated registration information and non-active registry status can have harsh consequences

[46]See Art 40, Art 29 I and II for a full description of the situations that can get the register suspended.

[47]Art 29 Inciso II

[48]Art 29 Inciso I

[49]Art 40 Par 2o.

[50]listed in item I of the caput of art. 29

on the entity according to the regulation. Thus, giving them actual incentives to comply with their registration responsibilities.

Inconsistent partners and administrators' data can prevent entities from closing their registries and end obligations along with the tax authority[51]. Additionally, an entity with an inapt registry loses the validity of its invoice on any tax effects in favor of an interested third party[52] and gets prevented from: transacting with banking establishments, including the handling of checking accounts, making financial investments and getting loans; obtaining fiscal and financial incentives; carrying out credit operations that involve the use of public resources; participating in public auctions to provide services for the government; and, entering into covenants, agreements, adjustments or contracts that involve public resources[53]. All of which corroborates in favor of the CNPJ being a reliable administrative data source to study formal entrepreneurship in Brazil.

The procedure to enroll or modify registration information in CNPJ also goes in the same direction. In order to perform any registration act[54], one must go through a procedure validated by RFB. First, the registration act request must be submitted online through a website administered by RFB[55]. Then, RFB makes available several documents related to registration information in the CNPJ for filling[56]. Once the documentation is filled out it must be submitted to RFB through that same website. RFB, then, screens the filled documentation against any data inconsistencies. It is only after going successfully through RFB's screening that the form for the registration act is made available to the requester. With that in hand, the form must be filled out and signed by the person responsible for the entity. The registration act is formalized once that form is delivered to RFB along with a list of additionally required documentation.

The responsible can change registration data in terms of the head office's corporate name; legal nature; social capital; the size of the company; the entity's representative in the CNPJ; to the representative; the partners and administrators data; to the responsible federative entity, in the case of Public Administration entities; bankruptcy; judicial recovery; intervention; the inventory of the individual entrepreneur or the holder of an individual real estate or limited

[51] *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 28.

[52] According to *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 48. the invoice from an inapt registry company loses validity on being: deducted as cost or expense, in determining the calculation basis for Corporate Income Tax (IRPJ) and Social Contribution on Net Income (CSLL); deducted in determining the basis for calculating the Personal Income Tax (IRPF); used as credit for the Non-cumulative Tax on Industrialized Products (IPI), the Contribution to PIS/Pasep and the Contribution to the Financing of Social Security (Cofins); used to justify any other deduction, reduction, reduction, compensation or exclusion related to the taxes administered by the RFB.

[53] *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 46

[54] *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 13 states that the following are registration acts in the CNPJ: registration; alteration of registration data and registration status; closing; reinstatement of registration; and declaration of nullity of the registration act.

[55] http://www.redesim.gov.br/

[56] *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art 14, the documents shared when a registration act request is first submitted are : Legal Entity Registration Form (FCPJ); the partners and administrators registration form; Specific Form of the agreement; Final beneficiaries file.

liability company; judicial or extrajudicial liquidation; incorporation; fusion; partial or total spin-off[57]. Importantly, although changes in legal nature and industry are possible, if made by an establishment participating in MEI out of the individual entrepreneur legal nature, into an industry not allowed in the program or opening a branch establishment, that establishment is automatically excluded from the program[58].

[57] *INSTRUÇÃO NORMATIVA RFB Nº 1863, DE 27 DE DEZEMBRO DE 2018*, Art. 17.

[58] *Resolução CGSN no 94/2011 de 29 de novembro de 2011*, Art. 105.

Table A6: Business entities - Concla/IBGE

Code	Title	Included
201-1	Empresa Pública	
203-8	Sociedade de Economia Mista	
204-6	Sociedade Anônima Aberta	x
205-4	Sociedade Anônima Fechada	x
206-2	Sociedade Empresária Limitada	x
207-0	Sociedade Empresária em Nome Coletivo	x
208-9	Sociedade Empresária em Comandita Simples	x
209-7	Sociedade Empresária em Comandita por Ações	x
212-7	Sociedade em Conta de Participação	
213-5	Empresário (Individual)	x
214-3	Cooperativa	
215-1	Consórcio de Sociedades	
216-0	Grupo de Sociedades	
217-8	Estabelecimento, no Brasil, de Sociedade Estrangeira	
219-4	Estabelecimento, no Brasil, de Empresa Binacional Argentino-Brasileira	
221-6	Empresa Domiciliada no Exterior	
222-4	Clube/Fundo de Investimento	
223-2	Sociedade Simples Pura	x
224-0	Sociedade Simples Limitada	x
225-9	Sociedade Simples em Nome Coletivo	x
226-7	Sociedade Simples em Comandita Simples	x
227-5	Empresa Binacional	
228-3	Consórcio de Empregadores	
229-1	Consórcio Simples	

Source: Legal nature table (2009 version) Concla/IBGE.
Notes: This table presents the business entities (1-digit aggregation code 2) from the legal nature table classification. It is compounded by the entity's type classification code, title and a last column indicating presence in the sample used in our analysis. In order to study entrepreunership accurately we keep only the private entities endowed with legal personality under private law, with profitable purposes and the individual entrepreneurs in our sample.

Table A7: CNPJ native registration information used in the analysis

Variable	Description
MEI	Indicator variable of participation in the MEI program
Opening date	Entity's start of activity date
Industry	CNAE/IBGE 7-digit industry code (2.0. version)*
State	State code
Legal nature	CONCLA/IBGE legal nature code
Corporate name	Legal name of the entity
Registration status	Status can be either Null, Active, Suspended, Inapt or Closed
Date of Registration status	Date of registration status

Table A8: Note: This table shows the native registration information of entities in CNPJ that we selected to use in our analysis.

MEI firms register in RAIS or CNPJ?

As discussed in the paper, we believe that the major reason for why Rocha et al. (2018) underestimate so much the effects of the program on formal firm activity is that microentrepreneurs are not obliged to report to RAIS. Unfortunately, at the time they wrote their paper, the CNPJ data set was not available.

Initially, it was unclear if the MEI with no employee would have to report the RAIS Negativa as it is required to the other firms in Brazil. The only clearly regulation exemption in the law was regarding social security contribution information to the relevant authorities. In 2010, several MEI were notified by the authorities charging a fine for not reporting the RAIS, but in April 2010 the Labor Ministry made an official statement exempting them from the charges. In February 2011, the Labor Ministry officially determined the exemption. In November 2011, a change to the MEI Law altered the eligibility cap and clearly stated the exemption from reporting the RAIS[59].

A.1 Creating a Panel of Firm Level Growth for Each Entrepreneur

Combining our three administrative data sets (CNPJ, CNE and RAIS) we create a panel that tracks firm size (measured as number of formal employees) of all microentreprises registered in the MEI tax system between 2009 and 2015 as well as other firms owned by microentrepreneurs that registered MEI firms. Here we detail how we created this data set.

The RAIS data set contains the universe of the formal labor market. From it, we can construct the number of formal employees at all formal establishments in Brazil. Establishments in RAIS are identified by the same unique identifier as in the CNPJ and CNE data sets. As

[59] Here and here are links explaining in portuguese this whole debacle.

we described in sections 2 and A, MEI firm owners that do not have an employee don't need to report RAIS information to the ministry of labor, but the ones that do employ one worker must report it. We thus input zero employees to MEI firms that are active in CNPJ data but are not present in RAIS.

We can thus observe for each MEI firm, the number of workers and wage bill in every year that the firm is active. This would however be only a first order approximation of firm growth because anecdotal evidence suggests that some entrepreneurs prefer to register a new firm ID when they grow out of the MEI program. If this anecdotal evidence represents a large share of the MEI firms that grow, only matching MEI firms to RAIS through their unique firm identifier would underestimate the growth of MEI firms.

To overcome this issue, we use our two other administrative data sets, CNE and CNPJ to recover ownership information of firms[60]. Table A9 summarizes our firm ownership data. Each column corresponds to a year of firms in RAIS. The first row shows the share of firms that we have information on who the owners are. For most years of the sample we have more than 90% of firm owners. We also see that the firms that we do not have information are on average much smaller than the ones that we do know who the owners are.

With our ownership information we organize our data set at the microentreprenuer level, instead of the microentreprise level. Thus we take into account both the MEI firm that she owns and other possible firms that she opens. If she opens a regular firm after her MEI firm, the new firm is taken into account as the firm size for that microentrepreneur. If she has more than one firm than we sum the sizes of these two firms. If the owned firm has more than one partner, we input the same number of employees to all partners, not adjusting by number of partners[61]. Similar procedures are made with the wage bill, which we define as the total amount of wages paid in December of the respective year.

[60] To our understanding, that was the same data procedure used in Colonnelli et al. (2020).

[61] One could argue that such adjustment is needed. For example, in a firm with 5 partners, one could think that we should divide the number of employees by five and input this share as the number of workers to the respective microentrepreneur. As our results point to the lack of growth, we believe we are conservative in our measure, and we are actually overestimating the number of employees under the supervision of thes microentrepreneurs.

Table A9: Summary Statistics from Ownership Data 2009-2018

	(1) 2009	(2) 2010	(3) 2011	(4) 2012	(5) 2013	(6) 2014	(7) 2015	(8) 2016	(9) 2017	(10) 2018
Sh. with Ownership Info	0.936	0.931	0.926	0.923	0.918	0.916	0.915	0.914	0.907	0.888
Sh. with CNE Own Info	0.784	0.777	0.769	0.765	0.755	0.751	0.742	0.734	0.714	0.670
Sh. with CNPJ Own Inco	0.702	0.696	0.691	0.685	0.678	0.671	0.672	0.674	0.674	0.672
Sh. With Own Info excluding RJ and MG	0.943	0.939	0.935	0.933	0.929	0.928	0.930	0.931	0.924	0.904
Sh. CNE not RJ or MG	0.850	0.845	0.840	0.836	0.828	0.826	0.820	0.815	0.796	0.747
Sh. Receita not RJ or MG	0.683	0.677	0.672	0.667	0.661	0.655	0.656	0.658	0.659	0.658
Avg. Wage Bill (all firms)	33489.5	34645.9	35836.9	36586.7	37321.1	36834.4	34182.1	33248.6	33793.7	34166.0
Avg. Number of Workers (all firms)	14.08	14.34	14.38	14.21	14.03	13.73	13.16	12.74	12.74	12.97
Avg. Wage bill (firms with Own. Info)	35394.9	36812.1	38256.5	39156.2	40156.8	39709.4	36862.6	35870.6	36707.3	37811.8
Avg. Wage bill (firms w/o Own Info)	5424.6	5446.4	5469.1	5622.3	5655.7	5624.2	5356.2	5317.9	5415.0	5281.6
Avg. Number of Workers (firms with Own. Info)	14.79	15.14	15.25	15.10	14.97	14.68	14.07	13.63	13.71	14.20
Avg. Number of Workers (firms w/o Own. Info)	3.633	3.581	3.558	3.548	3.452	3.411	3.356	3.263	3.254	3.230
Observations	1994586	2139009	2263606	2375698	2478617	2564489	2571444	2534032	2520734	2511241

Notes: This table shows summary statistics from the firm ownership data that we created. The data set was constructed by combining CNPJ, CNE and RAIS data described in section 2. The denominator is the total number of business establishments that hire at least one worker in RAIS at year t. A business establishment is defined by having *Natureza Juridica* between 2000 and 2999. Each column restricts the sample of business establishments for the ones that are active at year t where t is the year described in the top of the table. Wage bill is the average wage times the average number of workers.

The INSS data

The *Instituto Nacional do Seguro Social* (INSS) was established on June 27, 1990, through the merger of the Institute of Financial Administration of Social Security and the Assistance and the National Institute of Social Security. It is an autonomous agency that operates under the Ministry of Social Security and is responsible for recognizing entitlements and ensuring citizens' access to social security benefits, such as retirement, pensions, and maternity leave. We use two datasets from the INSS that are publicly available: the Beneficios Mantidos dataset and the *Anuário Estatístico da Previdência Social* (AEPS). In the following, we describe each dataset.

Beneficios Mantidos: Data on all benefits ever paid by the INSS. It includes active, terminated, and suspended benefits, in accordance with Decree No. 8,777/2016 and the Access to Information Act No. 12,527/2011. The data is extracted from the Unified Benefit Information System (SUIBE) and includes information on benefit types, medical condition codes (CID), beneficiary classification, gender, affiliation type, reason for termination or suspension, status group, municipality, state (UF), dependent relationships, beneficiary's date of birth, benefit termination date (DCB), benefit approval date (DDB), benefit start date (DIB), economic sector, benefit category, and monthly income. We follow the INSS classification to group beneficiaries by their affiliation type. The INSS categorizes affiliates into two main groups: "Employed Contributors" and "Other Contributors." Within the "Other Contributors" category, there is the Individual Contributor classification, which comprises entrepreneurs, self-employed workers, and those classified as equivalent to self-employed. Microentrepreneurs registered under the MEI program fall within this category.

AEPS: Social Security yearly reports on the system's figures in revenues and expenditures, the number of contributors, benefit flows and stock, and social security coverage. This data allows us to directly observe the revenues and the number of contributors from the MEI program, which we use to discuss the benefits of formalization to the government revenue. The INSS numbers reported are organized in two groups of contributors: Employed and Other Contributors. Employed contributors are the wage-dependent workers, except for domestic workers. Other contributors include the Individual Contributor, Domestic Worker, Facultative Contributor, and Special Insured. The Individual Contributor provides services of an urban or rural nature, on an occasional basis, to one or more companies, without an employment relationship; or engages in a remunerated economic activity of an urban or rural nature, whether for profit or not. It is the affiliation of entrepreneurs and self-employed. The Domestic Employee provides continuous service, with monthly remuneration, to a person or family, in a non-profit activity. The Special Insured is an individual who resides in a rural property or in a nearby urban or rural settlement and who, either individually or under a family-based economy, engages in agricultural activities on land of up to four fiscal modules; or who works as a rubber tapper or plant extractor, making these activities their primary means of livelihood; or as an artisanal fisherman, making fishing their regular profession or main source of income; or the spouse, partner, or child over 16 years old of the insured person who works with the family group. The Facultative Contributor is an individual over 16 years of age who joins the General Social Security System by making contributions.

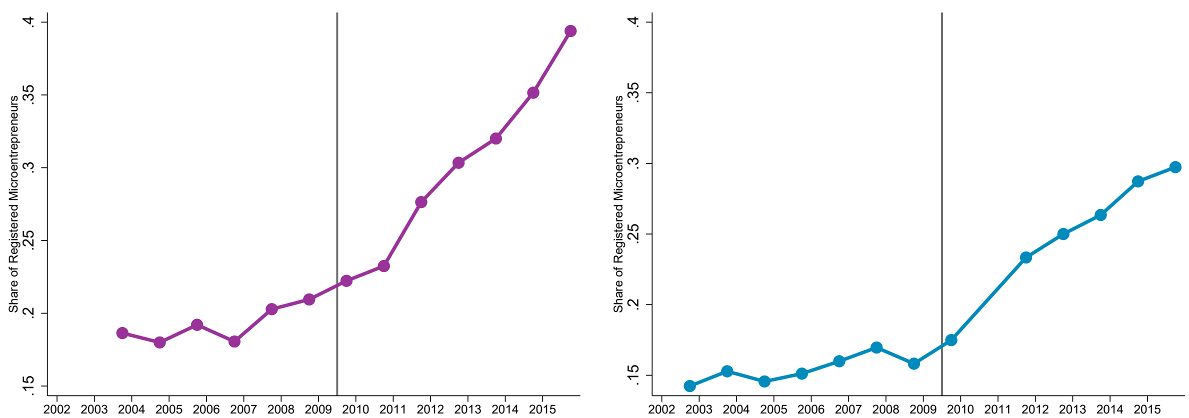
B Additional Results using PNAD and PME

In this appendix, we provide evidence using *Pesquisa Mensal do Emprego* to replicate some of the results we did using PNAD data. This is the data set used by Rocha et al. (2018) in their analysis[62]. It is a labor market survey collected montlhy in the largest metropolitan areas. It also comprises both informal and formal labor markets, but has a smaller sample than PNAD besides its difference in geographic representativeness.

We begin by replicating the share of microentrepreneurs that are formal in PME data. Our definition of formality is contributing to the social security network through the job[63]. We define a microentrepreneur as an individual who declares being self-employed. In PME we cannot restrict employers with at most one employee, as we did in PNAD, so replicate our PNAD findings using this slightly changed measure of microentrepreneurs

The share of formal microentrepreneurs in PME is extremely similar to the values in PNAD. We show in Figure A21 the share of formal microentrepreneurs between 2003 and 2015 in PME and PNAD using the same definition of microentrepreneur in both surveys. Values are higher in PME, which is expected given its geographical coverage of large urban areas, which, on average, have lower informality.

Figure A21: Share of Formal Microentrepreneurs PME and PNPAD



A. PME B. PNAD

Notes: All panels are built from PNAD data set. Sample is restricted to microentrepreneurs, which are defined as individuals that declare themselves self-employed or firm owners with at most one employee. Sample is restricted to individuals between 18 and 60 years. Being formal is defined as contributing to Social Security Network.

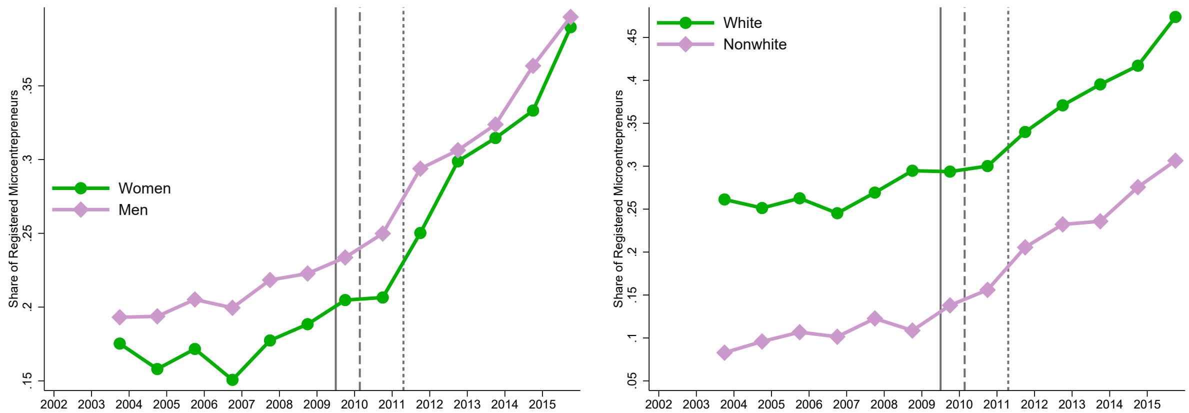
As a sanity check, we also do the heterogeneity by gender and race, variables that should be comparable across both surveys. We present them in Figure A22. We observe similar patterns

[62]Unfortunately, we could not find replication codes for the paper, thus we are going to provide analysis based on our understanding of the data and their results.

[63]In PME, this is equal to Variable V425==1

as to those observed in Figure A11 built using PNAD.

Figure A22: Share of Formal Microentrepreneurs by Group including Agriculture - PME survey



A. Gender B. Race

Notes: Both panels are built from PME data set from september of each year. Sample is restricted to microentrepreneurs, which are defined as individuals that declare themselves self-employed. Sample is restricted to individuals between 18 and 60 years. Being formal is defined as contributing to Social Security Network. Panel A divides microentrepreneurs iby gender. Panel D plots the share of microentrepreneurs years that are formal by race. Vertical lines represent the start of the different phases of the program.

C Characteristics of Microentrepreneurs that are Formalizing

Next, we use regression methods to further characterize the microentrepreneurs that are formalizing due to the program. We follow Abadie (2003) to identify the characteristics of compliers. In this setting, we call compliers the microentrepreneurs that are formal in the program's presence but would have been informal in its absence. Always Treated are the ones that are formal independently of the program, and the never treated are the microentrepreneurs that are informal with or without the existence of MEI program. To determine the characteristics $x_i \in X$ of compliers, we use PNAD data from 2002 to 2015 to estimate the following model by two stages least squares.

$$\begin{aligned} x_{it} \cdot Formal_{it} &= \alpha + \beta \cdot Formal_{it} + \delta \cdot trend_t + \epsilon_{it} \\ Formal_{it} &= \pi_0 + \pi_1 \cdot Z_{it} + \gamma \cdot trend_t + \eta_{it} \end{aligned} \tag{11}$$

This is simply an adaptation of the result described in Abadie (2003). We start from the model:

$$\begin{aligned} x_{it} \cdot Formal_{it} &= \alpha + \beta \cdot Formal_{it} + \delta \cdot trend_t + \epsilon_{it} \\ Formal_{it} &= \pi_0 + \pi_1 \cdot Z_{it} + \gamma \cdot trend_t + \eta_{it} \end{aligned}$$

For notation purposes, we will refer to the dummy $Formal_{it}$ as D_{it}. The population instrumental variable estimate of β is equal to:

$$\beta_{IV} = \frac{E[x_{it}D_{it}|z_{it}=1] - E[x_{it}D_{it}|z_{it}=0]}{E[D_{it}|z_{it}=1] - E[D_{it}|z_{it}=0]}$$

Now, let's describe what each of the objects in the expression represent:

$$E[x_{it}D_{it}|z_{it}=1] = E[x_{it}D_{it}|z_{it}=1, AT] \cdot Pr(AT) + E[x_{it}D_{it}|z_{it}=1, Comp] \cdot Pr(Comp)$$

Where AT represents Always Treated and Comp represents compliers. As we described before, in this setting we are calling compliers, the microentrepreneurs that are formal in the presence of the program, but would have been informal in its absence. Always Treated are the ones that are formal independent of the program and the never treated are the microentrepreneurs that are informal with or without the MEI program.

We know that both compliers and always treated groups are formal ($D_{it} = 1$) in the presence of the program ($z_{it} = 1$). Thus the first object can be written as:

$$E[x_{it}D_{it}|z_{it}=1] = E[x_{it}|AT] \cdot Pr(AT) + E[x_{it}|Comp] \cdot Pr(Comp)$$

Now, always treated are also always formal in the absence of the program. Thus the second object in the numerator of the β_{IV} estimate is:

$$\begin{aligned} E[x_{it}D_{it}|z_{it}=0] &= E[x_{it}D_{it}|z_{it}=0, AT]\cdot Pr(AT) \\ &= E[x_{it}|AT]\cdot Pr(AT) \end{aligned}$$

Thus, the numerator of the expression for β_{IV} is:

$$E[x_{it}D_{it}|z_{it}=1] - E[x_{it}D_{it}|z_{it}=0] = E[x_{it}|Comp]\cdot Pr(Comp)$$

Now, the denominator is exactly the probability of an individual being a complier:

$$E[D_{it}|z_{it}=1] - E[D_{it}|z_{it}=0] = Pr(Comp)$$

Which then yields our result:

$$\beta_{IV} = E[x_{it}|Comp]$$

Lastly, to estimate the difference in observable characteristic x between compliers and always treated we use the observed mean of characteristic x in the absence of the program:

$$Diff_{C-AT} = \hat{\beta_{IV}} - E[x_{it}|z_{it}=0, D_{it}=1]$$

Results: We restrict the sample to microentrepreneurs between 18 and 60 years old. As discussed in section 4, the industry variable in PNAD is not adequate to characterize the availability of the program. To overcome this, we use a procedure similar to Card and Giuliano (2016), where we use the time variation to determine the availability of the MEI program. Thus z_{it} is an indicator if the observation is in period $t \geq 2009$. The estimate of β yields the mean value of x for compliers. Furthermore, we can compute the mean value of x for always treated and compare the characteristics of compliers and always treated.

We present our estimates in Table A10. On the left panel of Table A10 we include PNAD 2009 and define the post-MEI period as years 2009-2015. However, PNAD 2009 was collected on September 2009, only three months after the program started, when it was still very small, as we have described in Section 3. For this reason, we also include estimates excluding 2009, where we define the post-MEI period as the surveys between 2011 and 2015[64].

In the first three rows of Table A10 we see that compliers are more likely to be women, nonwhite and younger. When we look at their education, the microentrepreneurs that became formal due to the program are more likely to have completed High School, but the share of them that have at least some college education is smaller compared to always-takers. In the next rows of Table A10 the characteristics x_{it} are dummies that indicate if microentrepreneur i was in the quintile q of the earnings distribution of microentrepreneurs in period t. We see that compliers are more likely to be in the first 3 quintiles of the earnings distribution than always takers.

[64] There was no PNAD collection in 2010

In particular, the largest share of compliers is in the 3rd quintile of the earnings distribution. Compliers are much less likely to be on the top of the earnings distribution than always-takers.

Table A10: Characteristics of Compliers and Differences relative to always takers

	Including 2009				Excluding 2009			
	All Formal Microent.		Compliers	Diff Compliers - Always Takers	All Formal Microent.		Compliers	Diff Compliers - Always Takers
	2002-2008 (Pre MEI)	2009-2015 (Post MEI)	(Newly Identified)		2002-2008 (Pre MEI)	2011-2015 (Post MEI)	(Newly Identified)	
Men	0.704	0.669	0.661	-0.0436	0.706	0.669	0.677	-0.0292
				(0.0281)				(0.0358)
Age	42.85	43.14	39.18	-3.676	42.77	43.14	38.18	-4.597
				(0.631)				(0.832)
White	0.729	0.620	0.514	-0.215	0.733	0.620	0.566	-0.167
				(0.0293)				(0.0365)
Less than H.S.	0.491	0.450	0.413	-0.0784	0.495	0.450	0.384	-0.110
				(0.0303)				(0.0390)
High School	0.252	0.301	0.393	0.141	0.248	0.301	0.439	0.190
				(0.0277)				(0.0362)
More than H.S.	0.257	0.250	0.194	-0.0627	0.257	0.250	0.177	-0.0800
				(0.0269)				(0.0345)
North or Northeast	0.128	0.182	0.266	0.138	0.125	0.182	0.277	0.152
				(0.0204)				(0.0264)
1st Earnings Quintile	0.0325	0.0591	0.0838	0.0512	0.0321	0.0591	0.0573	0.0252
				(0.0109)				(0.0131)
2nd Earnings Quintile	0.0874	0.121	0.193	0.106	0.0874	0.121	0.172	0.0843
				(0.0172)				(0.0208)
3rd Earnings Quintile	0.139	0.181	0.248	0.109	0.133	0.181	0.325	0.191
				(0.0211)				(0.0266)
4th Earnings Quintile	0.248	0.272	0.327	0.0790	0.252	0.272	0.236	-0.0154
				(0.0249)				(0.0302)
5th Earnings Quintile	0.493	0.367	0.148	-0.345	0.496	0.367	0.210	-0.285
				(0.0322)				(0.0374)
Weekly Hours Worked	46.52	43.93	44.91	-1.615	46.67	43.93	47.14	0.472
				(0.920)				(1.193)

Notes: This table shows the estimates from model 11. Sample consists of all microentrepreneurs between 18 and 60 years old between 2002 and 2015 in PNAD. On the right hand side panel we exclude 2009 from the sample. The compliers column shows the 2SLS estimate of the model specified in equation 11. In the column diff compliers - always takers, we subtract our 2SLS estimate by the observed characteristics from formal microentrepreneurs in the pre MEI period. Standard errors are in parenthesis.